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Initial Design Report

Traffic Light Control System

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1. Introduction and Interpretation

1.1 Problem Interpretation

The project requires the design and implementation of a **distributed real-time traffic-control system using QNX RTOS**. The network comprises five two-way roads (R1–R5), six signalized intersections (I1–I6), and a double-track railway corridor crossing R3–R5. The key challenge is to coordinate road and railway operation while preserving **local safety authority and independent operation**.

- **Local Intersection Control:** Each intersection (I1–I6) has a dedicated local controller that independently manages vehicle and pedestrian signals, responds to local sensor inputs and railway-crossing status, and continues safe operation if communication with Central is lost.
- **Central Monitoring and Command:** The Central controller maintains a real-time network view and provides a terminal-based operator interface. It may distribute timing/mode configurations and bound override requests, while all signal actuation remains under the authority of the relevant local controller.
- **Railway Coordination and Safety:** Each railway crossing (RC1–RC3) has a dedicated local controller responsible for its boom gates, road-facing flashers, and train signals. Intersection controllers react to crossing status but cannot command railway equipment. Train arrivals may occur at variable intervals, requiring railway pre-emption and adjacent-intersection coordination. Railway faults must locally force the affected train signal to STOP and be reported to Central.

1.2 System Assumptions

To bridge the gap between abstract requirements and concrete software implementation, specific operational parameters must be established. The following tables outline the foundational parameters guiding the system architecture, ensuring realistic traffic flow, correct safety behavior, and reliable inter-process communication.

Supporting calculations for numeric assumptions are provided in **Appendix B**.

1.2.1 Network and Road Usage Assumptions

Table 1. Network and Road Usage Assumptions

<i>ID</i>	<i>Assumption</i>	<i>Justification</i>	<i>Design Implication</i>
NU-01	Right-hand traffic (Vietnam convention): • median-side lane = through + permissive left turn; • kerbside lane = through + permissive right turn.	Vietnamese law requires right-hand traffic. The design uses the Mai Chí Thọ–Đồng Văn Cống intersection as a real-world reference, located in the former District 2, without reproducing its topology or measured timing. [1], [2], [3]	Lane layouts, sensor mappings, permitted movements, and conflict definitions follow right-hand traffic.
NU-02	Two lanes per direction: median side = through + permissive left; kerbside = through + permissive right. No dedicated turn lanes or protected turn-arrow phases are used in the Core design.	The brief requires two lanes per direction but makes turn lanes/arrows optional. Shared lanes preserve all turning movements while avoiding extra phases, per-lane sensing, and protected-turn complexity.	Each approach uses one circular vehicle signal head and approach-level sensing. The controller prevents prohibited phase conflicts; permissive turns remain subject to normal yielding rules.
NU-03	Four-approach intersection model: each approach has one vehicle signal head and one vehicle-presence sensor; each pedestrian crossing side has one push-button and one pedestrian signal head.	A symmetric layout supports one configurable controller design across all six intersections and aligns with the lecturer's clarification that four pedestrian buttons may be modelled per intersection.	Each L _x manages four vehicle signals, four vehicle-presence sensors, four pedestrian buttons, and four pedestrian signals.
NU-04	System boundary: minor side roads, private access roads, and	The brief allows unimportant side roads to be omitted so the model can focus on the specified network.	Unmodelled roads provide no sensor input and create no

	unsignalized approaches outside R1–R5 are excluded.		traffic demand or timing requirements.
NU-05	Road classification: R1–R2 are arterial roads along I1–I3–I5 and I2–I4–I6; R3–R5 are railway-crossing connectors linking I1–I2, I3–I4, and I5–I6 via RC1–RC3, respectively.	This classification distinguishes continuous roads suited to coordinated progression from shorter connectors affected by railway crossings.	R1–R2 receive arterial priority and green-wave coordination; R3–R5 use railway pre-emption and congestion-suppression logic.
NU-06	Fixed lane configuration: lane directions and permitted movements do not change dynamically or by time of day.	Dynamic lane reversal/reassignment is not required and would add control states unrelated to the assessed behavior.	Controllers may change signal timing and sequencing, but not lane directions or permitted movements.
NU-07	Sensing/control granularity: vehicles are sensed and signaled by approach, not by lane or turning movement; pedestrian inputs/signals operate separately by crossing side (NU-03).	Per-lane sensing and turn-specific signals are not required by the brief or the Core shared-lane design.	OFF_PEAK_SENSOR responds to approach-level demand and cannot distinguish a vehicle's intended movement.

1.2.2 Traffic Demand and Road Priority Assumptions

Table 2. Traffic Demand and Road Priority Assumptions

<u>ID</u>	<u>Assumption</u>	<u>Justification</u>	<u>Design Implication</u>
DP-01	Two normal demand modes: PEAK_FIXED and OFF_PEAK_SENSOR; no additional time-of-day demand tiers are modelled.	These cover the brief's fixed-timing and sensor-driven approaches. Advanced/updateable operation is provided through validated SET_MODE and SET_TIMING_PROFILE commands, not a third demand tier.	Normal operation uses only these two modes. Railway pre-emption, Central override, degraded operation, and fault handling remain separate overlay/safety states.
DP-02	Configurable Peak/Off-Peak boundary: the transition time is not fixed by the architecture.	The brief specifies time-of-day patterns but no exact transition time. Configurability also supports a compressed demonstration schedule.	C1, or the local clock when Central is unavailable, selects the applicable normal mode without a hard-coded transition hour.
DP-03	Sensor-based demand: no numeric vehicle-arrival rate or stochastic traffic-generation model is used; demand is represented by discrete sensor events.	The brief permits keyboard-simulated sensors and does not require traffic-flow simulation. Vehicle-arrival rates would neither affect controller decisions nor be independently verifiable.	Ordinary demand uses binary DEMAND_PRESENT/DEMAND_ABSENT states. The system neither counts vehicles nor estimates queue length; the separate advance detector only indicates whether a queue has reached its fixed detection point.
DP-04	Qualitative road priority: R1–R2 have higher base priority as arterials; R3–R5 have equal, lower base priority as connectors.	The brief requires demand/priority assumptions but provides no traffic counts or weights. The classification follows the network roles in NU-05 without inventing numeric demand.	PEAK_FIXED gives the arterial phase more green time; OFF_PEAK_SENSOR rests on the arterial phase when demand is absent. Connectors remain subject to railway pre-emption and congestion-suppression rules.
DP-05	Static road classification: arterial/connector priority does not change between Peak and Off-Peak operation.	Network role is independent of instantaneous sensor demand. Dynamic reclassification would reduce predictability without adding the required capability.	Sensor demand may affect phase selection or extension but never changes a road's arterial/connector classification.

DP-06	Bounded connector service: in <code>OFF_PEAK_SENSOR</code> , a latched connector demand is served after at most one opposing arterial service opportunity, subject to yellow and all-red clearance.	Arterial priority must not indefinitely starve a waiting connector approach.	While connector demand is pending, persistent arterial demand cannot repeatedly extend or immediately reselect the arterial phase. The phase-selection and extension-cap logic enforces this bound without numeric weighting.
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1.2.3 Traffic Light Timing Assumptions

Table 3. Traffic Light Timing Assumption

<u>ID</u>	<u>Assumption</u>	<u>Justification</u>	<u>Design Implication</u>
TL-01	Timing limits: minimum green = 8 s, maximum off-peak green = 40 s, yellow = 4 s, and all-red clearance = 2 s for each conflicting phase transition.	The brief provides no timing values. These tunable PoC values ensure bounded operation, with yellow and all-red intervals based on published signal-timing guidance. [4]	Every conflicting phase change completes yellow and all-red clearance before the next green; no mode, pre-emption, or Central command may bypass them.
TL-02	Peak fixed cycle: <code>PEAK_FIXED</code> uses a 90 s two-phase cycle: 48 s arterial green and 30 s connector green, each followed by the TL-01 clearance intervals.	This gives 54 s arterial and 36 s connector phase allocations, providing greater arterial service consistent with DP-04 . Railway pre-emption remains event-driven and independent of train headway. See Appendix B1 for the cycle derivation.	The cycle and green split remain fixed in <code>PEAK_FIXED</code> unless interrupted by a higher-priority safety or override state.
TL-03	Off-peak demand response: <code>OFF_PEAK_SENSOR</code> serves vehicle demand or latched pedestrian requests. Green extends in 4 second increments while demand persists, up to the TL-01 maximum; with no demand, the intersection rests on the arterial phase.	This implements sensor-driven operation while preserving the arterial default defined in DP-04 .	Phase selection and extension use bounded binary-demand logic while enforcing the DP-06 anti-starvation guarantee.
TL-04	Safe mode transition: changes between <code>PEAK_FIXED</code> and <code>OFF_PEAK_SENSOR</code> , whether clock-selected or requested by C1, take effect only at the next safe phase boundary.	This preserves minimum green and clearance requirements.	Mode changes remain pending until the current phase and required clearances complete.
TL-05	Pedestrian sequence: <code>WALK</code> → <code>FLASHING_DONT_WALK</code> → <code>DONT_WALK</code> , served concurrently with a compatible vehicle phase rather than a separate pedestrian-scramble phase.	The three indications and separation from conflicting vehicles align with published pedestrian-signal guidance, the brief does not require a scramble phase. [5]	<code>WALK</code> is permitted only with non-conflicting vehicle movements.
TL-06	Latched pedestrian request: one button press creates one pending request; repeated presses are coalesced. The request is served with its next compatible phase and has equal scheduling status to vehicle demand in <code>OFF_PEAK_SENSOR</code> .	Latching prevents lost requests, while coalescing prevents repeated presses from generating additional service requests.	Each crossing maintains one pending flag, cleared only after its <code>WALK</code> and clearance sequence completes.

1.2.4. Traffic Coordination Assumptions

Table 4. Traffic Coordination Assumptions

<u>ID</u>	<u>Assumption</u>	<u>Justification</u>	<u>Design Implication</u>
TC-01	Arterial coordination parameters: illustrative distances are I1-I3 = 350 m, I3-I5 = 400 m, I2-I4 = 320 m, and I4-I6 = 380 m, with an assumed travel speed of 60 km/h (16.67 m/s) .	The brief and lecturer clarification requires distance or travel time to inform coordination but not a physical survey. These are tunable design parameters, not measured values. The assumed speed is consistent with the current Vietnamese maximum for qualifying urban divided or one-way roads with at least two motor-vehicle lanes. [6]	These parameters determine the TC-02 arterial offsets; changing them requires recalculating the offsets, not modifying the coordination architecture.
TC-02	Green-wave coordination: applied only along I1-I3-I5 on R1 and I2-I4-I6 on R2, with each offset calculated as distance ÷ assumed travel speed.	These continuous arterial routes support platoon progression, while the shorter connectors are dominated by railway pre-emption and are not coordinated. See Appendix B2 for calculated offsets.	Only Lx sequencers on the two arterial chains apply these offsets; railway-controller timing is unaffected.
TC-03	Distributed offset control: C1 distributes arterial offsets via SET_TIMING_PROFILE; each Lx applies the parameter to its own sequencer.	This preserves local control: Central coordinates timing but never directly actuates signals.	The offset remains a validated timing parameter, not a new Central authority or local operating mode.
TC-04	Standalone fallback: if a valid coordination profile is unavailable or stale, Lx continues fixed or sensor-driven operation without waiting for C1.	Local operation must continue when Central or its communication link is unavailable.	Coordination remains a best-effort enhancement, never a dependency on safe local operation.
TC-05	Coordination recovery: after railway pre-emption disrupts the phase clock, the next valid timing profile restores the normal offset; no separate resynchronization state is used.	The existing timing-profile mechanism already restores coordination without an additional recovery process.	The controller resumes safe standalone cycling first, then reapplies coordination when a valid profile becomes available.

1.2.5 Congestion Control Assumptions

Table 5. Congestion Control Assumptions

<u>ID</u>	<u>Assumption</u>	<u>Justification</u>	<u>Design Implication</u>
CC-01	Advance queue detection: each crossing has one binary detector on each crossing-facing approach, approximately 60–80 m upstream of the stop line .	The upstream detector indicates when a queue reaches a defined threshold without vehicle counting or continuous queue-length estimation.	QUEUE_WARNING is used only for congestion suppression and recovery, not ordinary TL-03 phase extension.
CC-02	Crossing-closure suppression: from railway pre-emption until the crossing reports OPEN, movements toward the crossing receive no green. Compatible away-from-crossing and cross-traffic movements may be prioritized where geometry permits.	This follows the lecturer's instruction to stop feeding traffic toward a closed crossing without implying that signals physically reroute vehicles.	Suppression modifies scheduling within the current normal mode; it does not create a third traffic-demand mode.
CC-03	Post-reopening drain: the connector phase extends in 4 s increments while QUEUE_WARNING remains active, ending when the warning clears or the 60 s cap is reached.	This bounded rule clears backlog without inferring vehicle count or queue length from a binary detector. The 60 s cap is tunable. See Appendix B3 for the rule summary.	Repeated Boolean checks and the hard cap determine drain duration, after which normal phase selection resumes.

1.2.6. Railway Crossing Assumptions

Table 6. Railway Crossing Assumptions

<u>ID</u>	<u>Assumption</u>	<u>Justification</u>	<u>Design Implication</u>
RC-01	Crossing equipment: each crossing has two boom gates, road-facing flasher sets, train signals, train-approach sensors, and gate-position sensors. Two exit sensors are reserved for future enhancement and excluded from Core reopening logic.	The double-track corridor carries trains in both directions, requiring symmetric detection and protection.	The owner of RL _x exclusively controls gates, flashers, and train signals; other controllers receive status only.
RC-02	Independent crossings: RL1–RL3 independently manage RC1–RC3, with no shared cross-crossing state or train propagation between crossings.	The lecturer distinguishes railway from intersection controllers, and the topology requires no cross-crossing coordination.	Each RL _x broadcasts status only to its two adjacent L _x controllers; no L _x may command railway equipment.
RC-03	Train-approach warning: detection provides an approximately 45 s warning-to-arrival budget , covering flasher lead, gate movement, closed-confirmation margin, and adjacent-intersection clearance.	The brief provides no warning time. A component-based budget accounts for both gate operation and road-signal pre-emption, consistent with published crossing guidance. See Appendix B4 . [7]	This budget defines the required reaction window and, with RC-08 , the illustrative approach-sensor distance.
RC-04	Timed train occupancy: each simulated train occupies the crossing for 20 s after expected arrival. Gates remain closed until all active occupancy windows, including overlapping trains from either direction, have elapsed.	A fixed timer simplifies the Core demonstration without relying on live exit detection.	RL _x tracks overlapping occupancy windows and reopens only after all have expired.
RC-05	Timer-based reopening limitation: an abnormally slow or stalled train remaining beyond the 20 s occupancy window is not detected.	This explicitly documents the limitation of Core timer-based reopening and the unused exit sensors.	Exit-sensor confirmation is reserved for future enhancement and does not affect current reopening decisions.
RC-06	Gate-confirmed train authority: a train signal may show PROCEED only when the relevant gates are sensor-confirmed CLOSED ; elapsed closing time is never accepted as confirmation.	Timing estimates cannot verify the physical gate state; only position sensors establish the required safety condition.	Missing or contradictory confirmation forces STOP and raises a local fault.
RC-07	Common railway-controller logic: RL1–RL3 use identical control logic and differ only by configuration.	No source requirement justifies crossing-specific behavior.	One generic railway-controller implementation is deployed three times.
RC-08	Assumed train speed: maximum train speed is 80 km/h (22.2 m/s) .	The brief provides no train speed; this tunable planning assumption is used only to determine detection distance.	With the RC-03 warning budget, the illustrative detection distance is approximately 1,000 m ; see Appendix B4 .
RC-09	Central railway requests: C1 may send validated high-level requests to RL _x , such as fault clearance after repair, but never direct gate, flasher, or train-signal actuation commands.	This reconciles the brief's Central-command wording with the lecturer's requirement for local railway-equipment control.	RL _x validates each request and ACKs/NACKs it; unsafe or premature requests have no effect.
RC-10	Local railway fault response: on an equipment fault, RL _x independently forces the train signal to STOP and reports the fault to C1.	The safety response must not depend on Central connectivity or acknowledgement.	RL _x transitions to its local safe state without waiting for C1.

RC-11	Train-sensor fault detection: an approach input stuck active beyond its diagnostic timeout is detectable; a permanently silent input is not detectable without an independent health signal.	A binary input cannot distinguish "no train" from silent sensor failure.	Silent failure remains an accepted fail-open residual risk; sensor heartbeat/self-test is an out-of-Core mitigation.
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1.2.7. Pedestrian, Sensing and Abnormal Operation Assumptions

Table 7. Pedestrian Operation Assumptions

<u>ID</u>	<u>Assumption</u>	<u>Justification</u>	<u>Design Implication</u>
PA-01	Pedestrian scope: crossings are modelled only at I1-I6, not RC1-RC3.	The brief places pedestrian facilities at road intersections; railway-crossing pedestrian facilities are outside the selected scope.	Railway controllers have no pedestrian input or signals.
PA-02	Requests during railway pre-emption: pedestrian requests remain latched and are served in a compatible phase if available; otherwise, they wait until the conflicting restriction clears.	Railway pre-emption must take precedence without discarding pending pedestrian demand.	Request latches persist through railway events, while the conflict matrix determines when service is safe.
PA-03	Pedestrian button faults: a stuck-active input is coalesced into one pending request and reported as a fault; a permanently silent input is undetectable without an independent health signal.	A binary input can detect continuous assertion but cannot distinguish no button press from a dead input.	Stuck-active inputs cannot generate unlimited requests; silent failure remains an accepted limitation unless explicit self-test is added.

Table 8. Vehicle Sensing Assumptions

<u>ID</u>	<u>Assumption</u>	<u>Justification</u>	<u>Design Implication</u>
PA-04	Vehicle-presence sensing: each approach has one binary stop-line detector, used for phase selection and extension only in OFF_PEAK_SENSOR. The separate CC-01 detector is used only for congestion logic.	Approach-level demand is sufficient for the shared-lane design. Separating detector roles preserves fixed timing and avoids implying vehicle counting.	PEAK_FIXED is unaffected by ordinary presence demand; QUEUE_WARNING never substitutes for stop-line demand.
PA-05	Pedestrian-demand sensing: demand is generated only by push-button actuation; passive pedestrian detection is not modelled.	Push-button events provide deterministic, demonstrable input consistent with the brief.	PED_REQUEST originates only from the relevant button input.
PA-06	Vehicle-detector faults: a stuck-active detector is detectable and treated as continuous demand; a permanently inactive detector is indistinguishable from an empty approach without an independent health signal.	Treating prolonged inactivity as a fault could produce false alarms on genuinely quiet approaches.	Silent failure remains an accepted limitation; an explicitly reported or injected fault sets that approach to DEMAND_PRESENT.

Table 9. Abnormal and Failure Operation Assumptions

<u>ID</u>	<u>Assumption</u>	<u>Justification</u>	<u>Design Implication</u>
PA-07	Central-link failure: controllers send heartbeats every 1 s and declare the link unavailable after three consecutive misses . They retain the last validated timing parameters while local clocks continue selecting Peak/Off-Peak mode.	The brief requires continued local operation after communication or Central failure. The timeout is a tunable PoC diagnostic parameter.	DEGRADED_LOCAL disables coordination and override but does not freeze local mode selection or force all-red.

PA-08	Reconnection synchronization: on reconnection, the local controller reports its complete current state before accepting new Central commands.	Local state may have advanced while Central's view was stale.	Central resumes commanding only after state synchronization completes.
PA-09	Unsafe Central commands: any request that violates a safety invariant is rejected with NACK rather than applied or silently discarded.	Central may coordinate and request bounded overrides but cannot bypass local safety authority.	Each local command handler validates requests before state changes and returns ACK or NACK.
PA-10	Local fault containment: a controller or supervised-process fault affects only its own location, where the watchdog forces the documented safe outputs.	A local failure must not cascade through the distributed network.	FAULT_SAFE remains local: intersection outputs enter their safe road state; railway outputs hold gates down and train signals at STOP where actuation remains available.
PA-11	Every accepted REQUEST_OVERRIDE must specify a finite duration, capped at 5 minutes . The override expires automatically unless renewed through another validated request and may be cancelled earlier by Central.	The brief requires Central override commands for exceptional situations but specifies no duration. A bounded, auto-expiring override prevents a forgotten command from indefinitely suspending normal operation. The 5-minute cap is a tunable design parameter.[8]	Lx rejects a missing, non-positive, or over-limit duration with NACK. The timer begins when the override becomes active. On expiry or cancellation, Lx completes the required clearance intervals and returns to its previous normal mode.
PA-12	Each local controller returns ACK or NACK within 1 s of receiving a Central command. If execution must wait for a protected interval or safe phase boundary, the controller returns ACK_ACCEPTED_PENDING; only activation is deferred, and the request is revalidated before activation.	The brief requires Central-local command exchange but specifies no response deadline. A 1 s limit provides timely operator feedback without allowing Central commands to bypass local safety sequencing.	Command acknowledgement and command activation are treated separately; meeting the response deadline never requires immediate execution.

1.3 Target Grade Declaration

Our team targets a **Distinction-level delivery as the committed baseline**, with the system architecture intentionally designed to support progression toward **High Distinction**. The committed scope prioritizes a complete and reliable distributed traffic-control system with local autonomy, railway-safe operation, pedestrian handling, Central monitoring and control, and robust inter-controller communication. Higher-level capabilities, including coordinated traffic progression, congestion-aware control, and enhanced fault handling, will be incorporated progressively once the core system has been validated. This staged scope allows the team to pursue HD-level functionality without compromising the correctness, safety, or demonstrability of the complete system.

2. Physical Design and Layout

This section translates the abstract physical requirements into a mapped concrete environment. It establishes the real-world modeling and precise hardware layouts necessary to ground the distributed traffic control logic

2.1 Real-World Modeling

To ground the system design in a recognizable urban context, the physical layout is referenced against the traffic environment of *the Mai Chí Thọ - Đồng Văn Cống intersection* (part of the An Phu Intersection in Thu Duc City, Ho Chi Minh City), as shown in **Figure 1** below. The location provides a relevant Vietnamese urban-arterial context and supports the use of the right-hand traffic convention adopted in **NU-01**. It is used as a contextual reference rather than as a direct reproduction or quantitative model of the actual intersection.



Figure 1. Top view of the Mai Chí Thọ – Đồng Văn Cống intersection in Ho Chi Minh City [9]

The actual *Mai Chí Thọ intersection* has a substantially more complex multi-lane geometry than required by this project. The proposed system therefore abstracts the road environment into a uniform two-lane per direction model. The medium-side lane supports through and permissive left-turn movements, while the curbside lane supports through and permissive right-turn movements. This abstraction preserves the selected Vietnamese traffic convention while enabling a common local controller design to be deployed across all six intersections.

The project network further introduces the abstract R1–R5 road topology and a double-track railway corridor, neither of which is intended to reproduce the physical layout of the reference site. Inter-intersection distances of 320–400 m and advance queue-detector placement of 60–80 m are therefore treated as independently defined proof-of-concept parameters rather than measurements derived from *Mai Chí Thọ - Đồng Văn Cống*. These parameters provide the spatial basis required for the coordination and congestion-control logic defined in Sections 1.2.4 and 1.2.5.

2.2 Intersection Layout Diagram

To bridge the gap between the physical requirements and the software architecture, the following 2D layout diagrams establish the exact placement of all hardware peripherals managed by the local controllers. These layouts enforce the right-hand traffic convention and map directly to the system component nomenclature defined in **Appendix A**. To view these two diagrams in greater detail, please access the [A3_report_layoutdiagram](#).

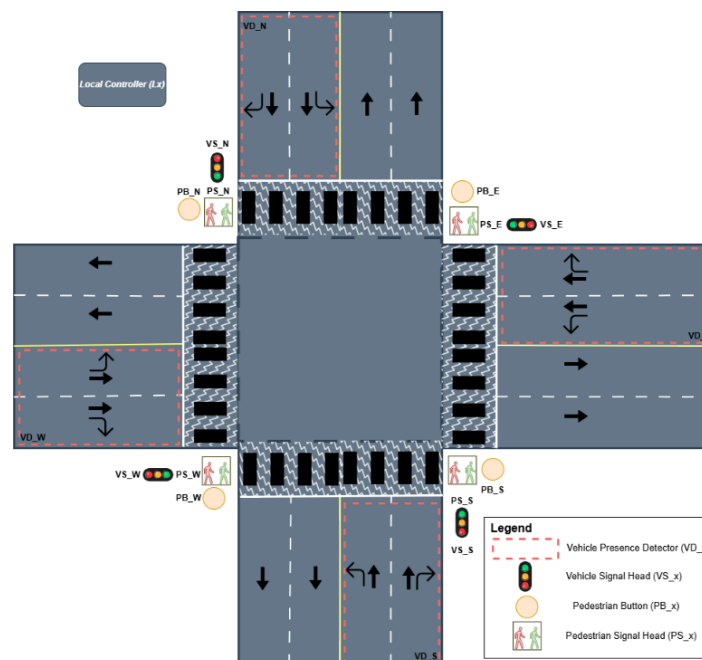


Figure 2. Physical Layout of a Standard Road Intersection (L1-L6)

The standard intersection utilizes a symmetrical 4-way layout, which is shown in **Figure 2**. Each of the four approaches is equipped with an approach level vehicle presence detector (VD_x) positioned behind the stop line. To optimize sidewalk space and consolidate wiring, the hardware on the driver's right-hand kerb is integrated into a single unified pole. This shared location mounts the standard vehicle signal head (VS_x) alongside the pedestrian signal head (PS_x), with the pedestrian push button (PB_x) conveniently located directly below the pedestrian light so individuals can easily request to cross the street.

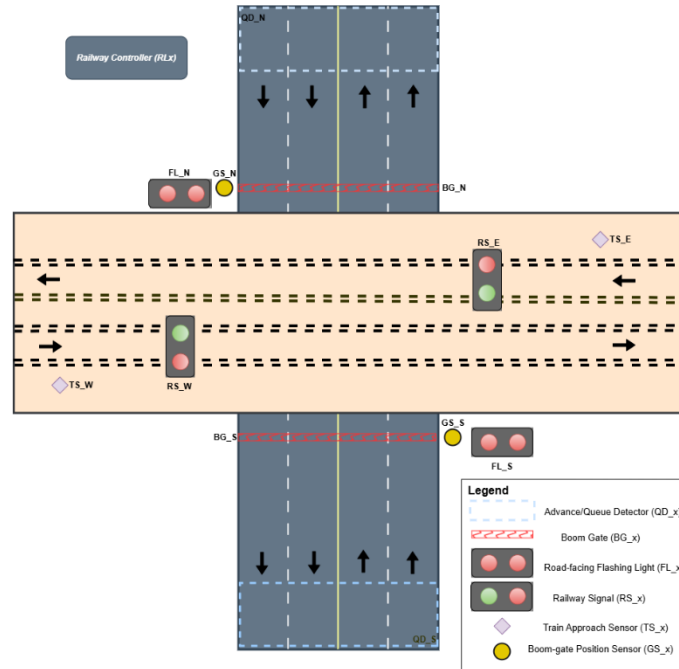


Figure 3. Physical Layout of a Double-Track Railway Crossing (RC1 – RC3)

The railway crossing layout illustrates a full barrier safety configuration, which is shown in **Figure 3**. When preemption is triggered by the upstream train approach sensors (TS_x), the crossing is secured by road facing flashing lights (FL_x) and two boom gates (BG_x) that extend across all lanes to prevent vehicle circumvention. Gate position sensors (GS_x) are located at the base of the barriers to confirm closure before the railway signals (RS_x) authorize a train to proceed. Additionally, advance queue detectors (QD_x) are placed 60 to 80 meters upstream on the approach lanes to support congestion suppression logic.

3. Functional Specifications

This section outlines how physical and logical actors interact with the system boundaries. It provides comprehensive Use Case diagrams and written scenarios to establish the externally observable behavior of the traffic network. To view these two use case diagrams in greater detail, please access the [A3_report_sequencediagram](#).

3.1 Use Case Diagrams

To effectively model the interactions within the distributed traffic-control system, it is essential to define the external entities, or actors, that interact with the system boundaries. These actors represent the physical users, hardware triggers, and cross-boundary systems that drive the flows:

- **Vehicle / road user:** Initiates demand for normal phases and requires congestion suppression and drain phases during railway pre-emption (*UC-01, UC-03, UC-05*).
- **Pedestrian:** Actuates pushbuttons to request safe crossing intervals (*UC-02*).
- **Train:** Activates track approach sensors, initiating high-priority railway protection sequences (*UC-04*).
- **Hardware Sensors & Inputs (Secondary):** Encompasses vehicle-presence detectors, pedestrian pushbuttons, train-approach sensors, gate-position sensors, queue detectors, and fault inputs that directly trigger local physical state changes (*UC-01, UC-02, UC-04, UC-05, UC-06*).
- **Central Control Room Operator:** A human user who configures operating parameters, applies clear-route overrides, and monitors network status and faults from the terminal (*UC-03, UC-06, UC-07, UC-08, UC-09, UC-10*).

- **Communication Network / Controllers:** The Central Controller and Local Controllers act as cross-boundary actors to each other, passing telemetry, sending IPC commands, or triggering fallback modes upon link loss (*UC-10*).

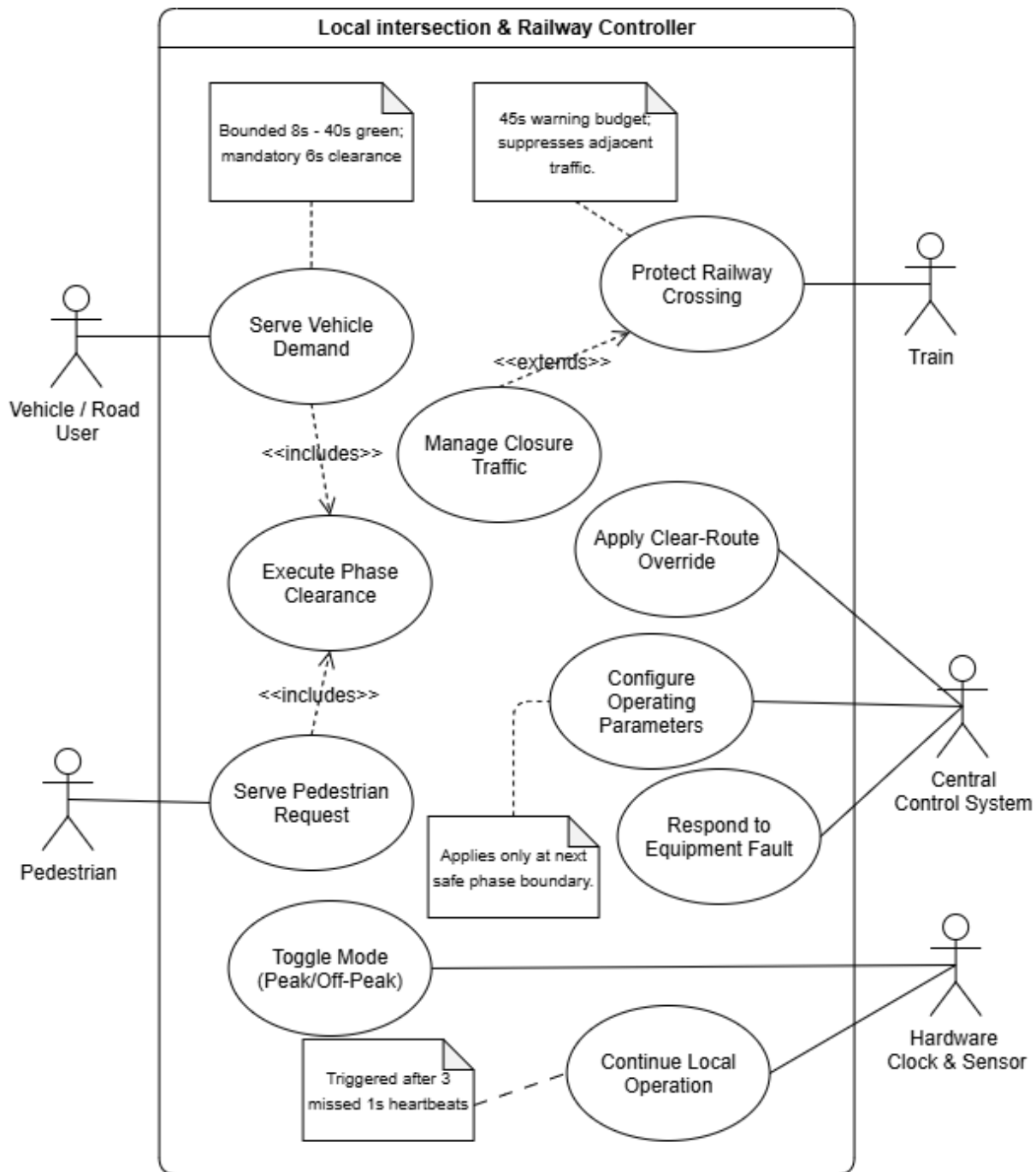


Figure 4. UML Use Case Diagram for Local Node Operations

The Local Node Operations, as illustrated in **Figure 4**, are encapsulated by the autonomous functions of the Local Intersection and Railway Controllers. Within this boundary, physical actors such as Vehicles, Pedestrians, and Trains trigger localized and safety critical use cases like serving demand, managing closure traffic, and protecting railway crossings. A fundamental characteristic of this local autonomy is the `<<includes>>` relationship linking demand service to the mandatory execution of phase clearances, which ensures that hardcoded safety invariants, such as yellow and all red intervals, are rigorously enforced and never bypassed. Furthermore, the diagram explicitly models the Central Control System as an external actor that pushes operating parameters and clears route overrides down to the local node. It also introduces the fallback use case to continue local operation triggered by the hardware clock, highlighting the decentralized failsafe architecture where local controllers independently validate remote commands and handle communication loss.

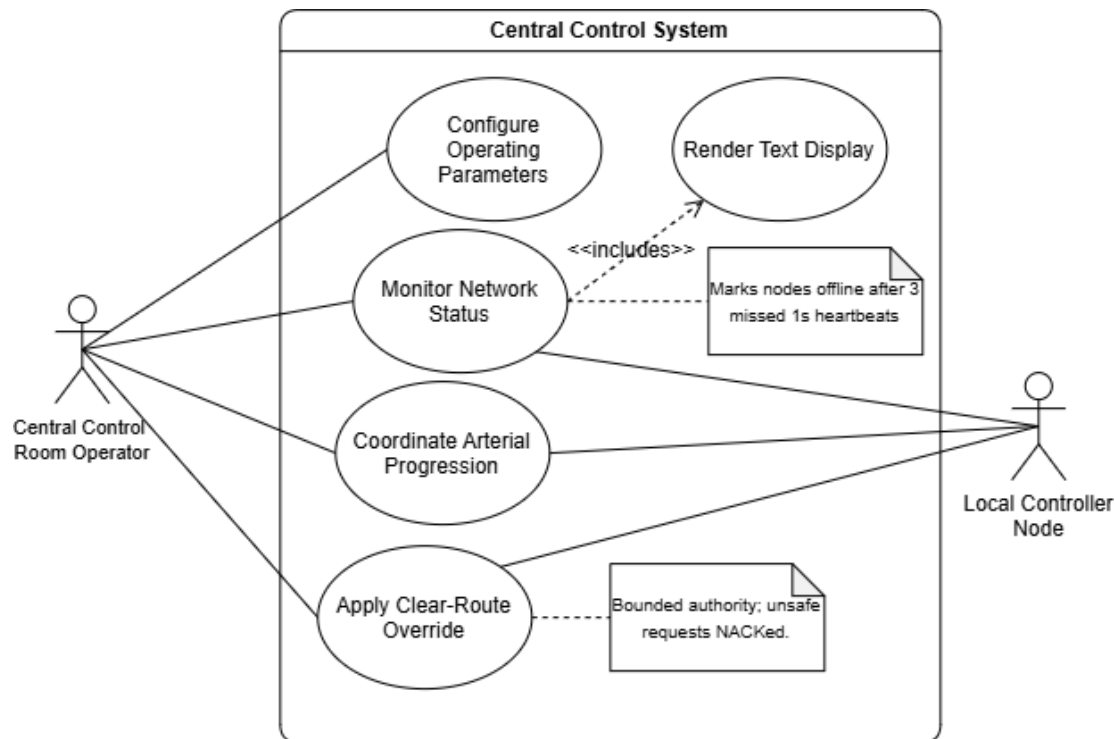


Figure 5. UML Use Case Diagram for Central Control Operations

Depicting the Central Control Operations, **Figure 5** focuses on the supervisory and coordination capabilities of the central control room. Here, the Control Operator serves as the primary human actor interacting with the system boundaries to “*Coordinate Arterial Progression*”, “*Configure Operating Parameters*”, and “*Apply Clear Route Overrides*”. The “*Monitor Network Status*” use case features an *<<includes>>* relationship with “*Render Text Display*”, demonstrating that telemetry received from the local controller nodes must be intuitively formatted as simple text messages for the operator. Additionally, the diagram illustrates how network monitoring inherently manages link failures and equipment faults, as denoted by the attached requirement to mark nodes offline after three missed heartbeats, ensuring robust status alerting even during partial network outages.

3.2 Scenarios and Written Specifications

The following use cases define the principal externally observable behaviors of the distributed traffic-control system. They progress from normal traffic operation through coordination, railway protection, Central control functions, and degraded operation. Each Main Flow describes a complete successful outcome, while each alternative flow identifies its branching step and resulting endpoint.

3.2.1 UC-01 - Serve Vehicle Demand

Table 10. Use Case 01 Specification

Field	Description
Name	UC-01 – Serve Vehicle Demand
Goal	Serve arterial and connector traffic safely according to the active normal operating mode.
Summary	The system operates each intersection using fixed phase timing during Peak periods and detected demand during Off-Peak periods. All phase changes preserve the configured green and clearance limits.
Primary Actors	Vehicle / road user
Secondary Actors	Vehicle-presence sensor
Pre-conditions	1. The intersection is operating in PEAK_FIXED or OFF_PEAK_SENSOR. 2. No local fault, railway restriction, or Central override prevents the requested movement. 3. The current timing profile has been locally validated.

Triggers	A scheduled phase becomes due in <code>PEAK_FIXED</code> , or vehicle demand becomes present in <code>OFF_PEAK_SENSOR</code> .
Main Flow	1. The system identifies the active normal operating mode. 2. The system selects the next eligible vehicle phase using the fixed schedule or recorded demand, as applicable. 3. If a conflicting phase is active, the system completes its required minimum-green interval. 4. The active phase passes through 4 seconds of yellow and 2 seconds of all-red clearance. 5. The selected vehicle phase receives green for at least 8 seconds. 6. In <code>OFF_PEAK_SENSOR</code>, the system checks demand every 4 seconds and extends green while demand persists, up to 40 seconds. 7. When service is complete, the system safely terminates the phase through the required clearance intervals. 8. The served demand is cleared, and the system returns to normal phase selection.
Alternative Flows	2.1 Peak Fixed operation: 2.1.1. The system selects the next phase from the fixed 90-second cycle. 2.1.2. Ordinary presence sensors do not alter the fixed phase duration. 2.1.3. Continue at Step 3. 2.2 No Off-Peak demand is present: 2.2.1. The system retains or returns to the arterial resting phase. 2.2.2. The system waits for new vehicle or pedestrian demand. 2.2.3. The use case ends in the normal arterial rest state. 3.1 A higher-priority condition arises: 3.1.1. The system preserves all mandatory clearance intervals. 3.1.2. The applicable railway, override, or fault use case takes control. 3.1.3. The use case ends with the vehicle demand served or safely retained as pending. 6.1 Demand clears before the Off-Peak maximum: 6.1.1. The system completes the current 4-second extension increment. 6.1.2. Continue at Step 7. 6.2 Connector demand is waiting: 6.2.1. The system prevents repeated arterial extension or immediate arterial reselection. 6.2.2. The connector demand is served after no more than one opposing arterial service opportunity. 6.2.3. Continue at Step 3 for the connector phase.
Post-conditions	1. The selected vehicle demand has been served or remains safely pending under a higher-priority restriction. 2. No conflicting movements have received green simultaneously. 3. The intersection has resumed normal phase selection or its arterial rest state.
Business Rules	BR-1: Every conflicting transition must include 4 seconds of yellow and 2 seconds of all-red clearance (TL-01). BR-2: Off-Peak green must remain between 8 and 40 seconds and may extend only in 4-second increments (TL-01, TL-03). BR-3: With no demand, Off-Peak operation rests on the arterial phase (DP-04, TL-03). BR-4: A latched connector demand must not be indefinitely starved (DP-06). BR-5: Ordinary presence demand affects phase selection only in <code>OFF_PEAK_SENSOR</code> (PA-04).

3.2.2 UC-02 - Serve Pedestrian Request

Table 11. Use Case 02 Specification

Field	Description
Name	UC-02 – Serve Pedestrian Request
Goal	Complete a requested pedestrian crossing safely with a compatible vehicle phase.

Summary	A pedestrian button press creates one pending request. The system serves it with the next compatible vehicle phase and completes the full pedestrian clearance sequence before releasing conflicting traffic.
Primary Actors	Pedestrian
Secondary Actors	Pedestrian push-button
Pre-conditions	1. The pedestrian crossing is located at one of I1–I6. 2. The relevant pedestrian signal displays DONT_WALK. 3. No pedestrian service is currently active for that crossing side.
Triggers	The pedestrian presses the relevant push-button.
Main Flow	1. The pedestrian presses the crossing button. 2. The system detects the button event and latches one pending request. 3. The system identifies the next vehicle phase compatible with the requested crossing. 4. When the compatible phase begins, the pedestrian signal changes to WALK. 5. After the WALK interval, the signal changes to FLASHING_DONT_WALK for clearance. 6. The signal returns to DONT_WALK before any conflicting vehicle movement receives green. 7. The system clears the pending request. 8. The pedestrian service is complete, and ordinary phase operation continues.
Alternative Flows	2.1 A request is already pending: 2.1.1. The system coalesces the new press into the existing request. 2.1.2. No additional service request is created. 2.1.3. Continue at Step 3. 2.2 Button remains stuck active: 2.2.1. The input is coalesced into one request. 2.2.2. The system reports the stuck-active input as a fault. 2.2.3. Continue at Step 3 without generating additional requests. 3.1 Railway restrictions prevent immediate service: 3.1.1. The request remains latched. 3.1.2. If another compatible phase is available, continue at Step 4 with that phase. 3.1.3. Otherwise, wait until the restriction clears and then return to Step 3. 5.1 A Central override arrives during pedestrian clearance: 5.1.1. The system does not truncate WALK or FLASHING_DONT_WALK. 5.1.2. The override waits or is rejected according to its safety constraints. 5.1.3. Continue at Step 6.
Post-conditions	1. After successful service, the pedestrian signal displays DONT_WALK and the request is cleared. 2. If service remains restricted, the request stays latched. 3. No conflicting vehicle movement has received green before pedestrian clearance completed.
Business Rules	BR-1: Pedestrian service must follow WALK→FLASHING_DONT_WALK→DONT_WALK (TL-05). BR-2: WALK may run only with a non-conflicting vehicle phase (TL-05). BR-3: Repeated presses create no more than one pending request (TL-06). BR-4: A request received during railway pre-emption must not be discarded (PA-02). BR-5: Pedestrian facilities are modelled at intersections only (PA-01).

3.2.3 UC-03 - Coordinate Arterial Progression

Table 12. Use Case 03 Specification

Field	Description
Name	UC-03 – Coordinate Arterial Progression
Goal	Coordinate successive arterial green phases using validated travel-time offsets.
Summary	The system applies distance-and-speed-based timing offsets along R1 and R2 to support vehicle progression between successive intersections. Coordination remains an enhancement rather than a dependency for safe local operation.
Primary Actors	Central Control Room Operator

Secondary Actors	None
Pre-conditions	1. A valid arterial coordination profile is available. 2. The relevant intersections are operating normally. 3. No railway or fault state is currently disrupting the coordinated phases.
Triggers	The operator submits a validated arterial coordination profile for an arterial chain.
Main Flow	1. The operator submits the validated arterial coordination profile to Central. 2. Central supplies each relevant intersection with its assigned timing offset. 3. Each local controller validates the received offset. 4. Each controller applies its offset at a safe phase boundary. 5. The first intersection in the arterial chain begins its coordinated green phase. 6. Downstream intersections begin their corresponding green phases according to the configured travel-time offsets. 7. Vehicles travelling near the assumed arterial speed receive the intended green-progression opportunity. 8. Each controller reports its active profile and phase state to Central. 9. The coordinated profile remains active while valid, completing the use case in normal coordinated operation.
Alternative Flows	3.1 An offset violates a local safety bound: 3.1.1. The receiving controller rejects the profile with NACK. 3.1.2. The controller retains its previous valid configuration. 3.1.3. The use case ends with that intersection operating safely without the rejected profile. 2.1 The coordination profile is missing or stale: 2.1.1. The affected controller does not wait for Central. 2.1.2. It continues standalone fixed or sensor-driven operation. 2.1.3. The use case ends in safe standalone operation. 6.1 Railway pre-emption disrupts a coordinated intersection: 6.1.1. The affected controller suspends the conflicting coordinated movement and follows the railway restriction. 6.1.2. After the restriction clears, it resumes safe standalone cycling. 6.1.3. The next valid timing profile restores ordinary coordination without a separate resynchronization state. 6.1.4. The use case ends when coordinated or safe standalone operation resumes.
Post-conditions	1. The arterial chain operates with validated offsets, or its intersections remain safe in standalone operation. 2. Every local controller retains authority over its own phase timing. 3. Coordination has not bypassed any local safety rule.
Business Rules	BR-1: Coordination applies only to I1-I3-I5 on R1 and I2-I4-I6 on R2 (TC-02). BR-2: Offsets are derived from distance divided by assumed travel speed (TC-01, TC-02). BR-3: Central distributes profiles, while each local controller validates and applies its own parameters (TC-03). BR-4: Missing or stale coordination must degrade to standalone operation (TC-04). BR-5: Railway disruption is recovered through the normal timing-profile path (TC-05).

3.2.4 UC-04 - Protect Railway Crossing

Table 13. Use Case 04 Specification

Field	Description
Name	UC-04 – Protect Railway Crossing
Goal	Secure the railway crossing before authorizing an approaching train to proceed.
Summary	When an approaching train is detected, the system warns road users, suppresses traffic toward the crossing, closes and verifies the gates, and maintains protection until every active train-occupancy window has elapsed.

Primary Actors	Train
Secondary Actors	Train-approach sensor, boom-gate position sensors
Pre-conditions	1. The crossing is OPEN. 2. Its train signals display STOP. 3. Its local railway controller is operational. 4. No unresolved crossing fault is active.
Triggers	A train-approach sensor reports TRAIN_APPROACHING.
Main Flow	1. The system detects the approaching train and registers its expected arrival and occupancy window. 2. The crossing enters WARNING, and the road-facing flashers activate. 3. The system reports the crossing state to the two adjacent intersections and Central. 4. Each adjacent intersection safely clears the movement heading toward the crossing and holds it at red. 5. After the flash-only warning interval, both boom gates begin closing. 6. The gate-position sensors confirm that the required gates are CLOSED. 7. The relevant train signal changes to PROCEED. 8. The gates remain closed and the crossing remains protected throughout the train's occupancy window. 9. After every active occupancy window has elapsed, the train signal returns to STOP. 10. The gates reopen, the flashers stop, and the crossing returns to OPEN. 11. The system reports OPEN to the adjacent intersections and Central, completing the crossing sequence.
Alternative Flows	3.1 Central communication is unavailable: 3.1.1. The local crossing sequence continues without Central dependency. 3.1.2. Status reporting resumes when communication becomes available. 3.1.3. Continue at Step 4. 6.1 Gate closure is not confirmed: 6.1.1. The train signal remains at STOP. 6.1.2. The crossing enters FAULT and holds the applicable road restrictions. 6.1.3. The system reports the fault to Central independently of the local safety response. 6.1.4. The normal crossing sequence terminates in the safe fault state. 8.1 A second train is detected before reopening: 8.1.1. The system registers an additional occupancy window. 8.1.2. The gates remain closed while either occupancy window remains active. 8.1.3. Return to Step 9 only after all active windows have elapsed. 8.2 Train-approach input remains active beyond its diagnostic timeout: 8.2.1. The system identifies a stuck-active sensor fault. 8.2.2. The crossing remains on the safe side with the gates down and train signal at STOP. 8.2.3. The fault is reported and the use case ends in the safe fault state.
Post-conditions	1. The crossing has returned to OPEN; or 2. The crossing remains in a documented safe fault state with the train signal at STOP and road access restricted. 3. No gate has reopened while an occupancy window remained active.
Business Rules	BR-1: Railway warning uses an approximately 45-second warning-to-arrival budget (RC-03). BR-2: A train signal may show PROCEED only after gate closure is sensor-confirmed (RC-06). BR-3: Gates must remain closed until every active occupancy window has elapsed (RC-04). BR-4: Only the owning railway controller may actuate railway equipment (RC-01, RC-02). BR-5: Railway protection must not depend on Central connectivity (RC-10). BR-6: A permanently silent approach sensor is not detectable by the Core design (RC-11).

3.2.5 UC-05 - Manage Closure Traffic

Table 14. Use Case 05 Specification

Field	Description
Name	UC-05 – Manage Closure Traffic
Goal	Prevent traffic from being fed toward an unavailable crossing and safely clear the queue after reopening.
Summary	Adjacent intersections suppress the movement toward a closing or closed crossing. After reopening, a bounded drain phase is used when the advance detector reports that the queue has reached its threshold.
Primary Actors	Vehicle / road user
Secondary Actors	Crossing-status input, advance/queue detector
Pre-conditions	1. The intersection is adjacent to a railway crossing. 2. It receives the current state of that crossing. 3. Its normal phase sequencer is operational.
Triggers	The adjacent crossing reports a state other than OPEN.
Main Flow	1. The system receives the non-open crossing status. 2. If the toward-crossing movement is active, the system completes its minimum-green requirement. 3. The movement safely transitions through yellow and all-red to red. 4. The system withholds green from the toward-crossing movement while the crossing remains unavailable. 5. Compatible cross-traffic and away-from-crossing movements continue where permitted. 6. The crossing reports OPEN while the advance detector reports QUEUE_WARNING. 7. The system starts the connector drain phase. 8. The system checks QUEUE_WARNING every 4 seconds and extends the drain phase while the warning remains active. 9. The warning clears before the 60-second cap. 10. The system safely terminates the drain phase and resumes ordinary phase selection.
Alternative Flows	4.1 Crossing enters FAULT: 4.1.1. The movement toward the crossing remains red. 4.1.2. Compatible non-conflicting traffic continues where safe. 4.1.3. The restriction remains until the fault clears and the crossing reports OPEN. 4.1.4. Return to Step 6 after reopening. 6.1 No queue warning exists after reopening: 6.1.1. The system skips the extended drain phase. 6.1.2. Normal phase selection resumes. 6.1.3. The use case ends in normal operation. 9.1 Queue warning remains active at 60 seconds: 9.1.1. The drain phase terminates at the hard cap. 9.1.2. The system completes the required clearance intervals. 9.1.3. Normal bounded phase selection resumes, ending the use case.
Post-conditions	1. Traffic toward an unavailable crossing remains suppressed. 2. After reopening, the drain phase has completed, reached its cap, or was unnecessary. 3. The intersection has resumed ordinary phase selection.
Business Rules	BR-1: A toward-crossing movement must receive no green until the crossing reports OPEN (CC-02). BR-2: The queue detector reports only whether the queue has reached its fixed detection point (CC-01). BR-3: Drain extension uses 4-second increments and a 60-second cap (CC-03). BR-4: Railway suppression may not bypass yellow or all-red clearance (TL-01).

3.2.6 UC-06 - Respond Equipment Fault

Table 15. Use Case 06 Specification

Field	Description
Name	UC-06 – Respond to Equipment Fault
Goal	Contain a railway equipment fault locally and expose it to the Central Control Room.
Summary	The system responds locally when a required railway-equipment state cannot be confirmed. The train signal is forced to STOP independently of Central communication, and the fault is reported for operator's attention.
Primary Actors	Hardware Sensors & Inputs
Secondary Actors	Central Control Room Operator
Pre-conditions	1. A railway controller is performing or supervising a safety-relevant operation. 2. The applicable equipment state is being monitored.
Triggers	A required gate-state confirmation is missing, contradictory, or explicitly reports a fault.
Main Flow	1. The system detects that the expected safe equipment state cannot be confirmed. 2. The crossing enters FAULT. 3. The train signal is immediately forced or held at STOP. 4. The gates, flashers, and adjacent road restrictions are held in their applicable safe states. 5. Independently of the local safety response, the system reports the fault to Central. 6. Central displays the fault to the operator. 7. The fault remains latched and normal crossing operation remains suspended. 8. The use case ends with the crossing contained in its documented safe fault state.
Alternative Flows	5.1 Central communication is unavailable: 5.1.1. The local safe state remains active without waiting for Central. 5.1.2. The operator notification remains unavailable until reporting can resume. 5.1.3. Continue at Step 7. 7.1 Operator requests fault clearance after repair: 7.1.1. The local railway controller verifies that the physical fault has cleared. 7.1.2. If verification succeeds, it accepts the request and safely restores normal crossing readiness. 7.1.3. The use case ends with the crossing ready for normal operation. 7.2 Fault-clear request is premature: 7.2.1. The local controller detects that the unsafe condition remains. 7.2.2. It rejects the request with NACK. 7.2.3. The crossing remains in FAULT, and the use case ends in the safe fault state.
Post-conditions	1. The crossing remains protected with the train signal at STOP; or 2. After verified repair, the fault is cleared and the crossing is ready for normal operation. 3. No Central acknowledgement was required to establish the local safe state.
Business Rules	BR-1: Missing or contradictory gate confirmation must force STOP and raise a local fault (RC-06). BR-2: The local safety response and Central report are independent actions (RC-10). BR-3: Central may request fault clearance but may not actuate railway equipment (RC-09). BR-4: An unsafe request must be rejected with NACK (PA-09). BR-5: A local fault must not be cascaded to unrelated locations (PA-10).

3.2.7 UC-07 - Configure Operating Parameters

Table 16. Use Case 07 Specification

Field	Description
Name	UC-07 – Configure Operating Parameters
Goal	Apply a valid mode or timing-profile change without disrupting safe active operation.
Summary	The operator requests a normal operating-mode or timing-profile change. The target controller validates the request and applies an accepted change only at a safe phase boundary.

Primary Actors	Central Control Room Operator
Secondary Actors	None
Pre-conditions	1. Central communication is available. 2. The target controller is operational and reporting its current state. 3. Central holds a valid target and proposed parameter set.
Triggers	The operator submits a SET MODE or SET TIMING PROFILE request.
Main Flow	1. The operator selects a target controller and enters the requested mode or timing parameters. 2. The operator submits the request. 3. The target controller validates the request against supported ranges and local safety constraints. 4. The controller returns ACK to indicate acceptance. 5. If a phase is active, the accepted change remains pending until the current phase and required clearances complete. 6. The controller applies the change at the next safe phase boundary. 7. The controller reports the updated mode or profile to Central. 8. Central displays the confirmed configuration, and the use case ends successfully.
Alternative Flows	3.1 Request violates a safety constraint: 3.1.1. The controller rejects the request with NACK. 3.1.2. Central displays the rejection reason. 3.1.3. The current validated configuration remains active. 3.1.4. The use case ends without applying the requested change. 3.2 Communication fails before acceptance: 3.2.1. No unconfirmed change is applied. 3.2.2. The controller continues using its last validated parameters. 3.2.3. Central marks the request as unsuccessful or unresolved. 3.2.4. The use case ends in autonomous local operation. 5.1 A higher-priority state is active: 5.1.1. The accepted change remains pending or is rejected as applicable. 5.1.2. Higher-priority behavior proceeds first. 5.1.3. If the request remains valid afterward, return to Step 6 . Otherwise, end without applying it.
Post-conditions	1. The accepted configuration is active and reported; or 2. The previous validated configuration remains active after rejection or communication failure. 3. No active phase or clearance interval has been unsafely truncated.
Business Rules	BR-1: Normal mode selection uses only PEAK_FIXED and OFF_PEAK_SENSOR (DP-01). BR-2: A mode switch takes effect only at a safe phase boundary (TL-04). BR-3: Timing values must preserve the configured green and clearance limits (TL-01). BR-4: Each local controller validates and applies its own profile (TC-03). BR-5: A request violating a safety invariant must be rejected with NACK (PA-09).

3.2.8 UC-08 - Apply Clear-Route Override

Table 17. Use Case 08 Specification

Field	Description
Name	UC-08 – Apply Clear-Route Override
Goal	Temporarily prioritize a requested through-movement without overriding local safety authority.
Summary	The operator requests a bounded clear-route override for a target intersection. The local controller accepts the request only if the movement can be served without violating railway protection, pedestrian clearance, or signal-transition requirements.
Primary Actors	Central Control Room Operator
Secondary Actors	None
Pre-conditions	1. Central communication is available.

	2. The target controller is operational and reporting on its current state. 3. The operator has selected a valid target movement and bounded duration.
Triggers	The operator submits <code>REQUEST_OVERRIDE(CLEAR_ROUTE, target, duration)</code> .
Main Flow	1. The operator selects the target through-movement and override duration. 2. The operator submits the override request. 3. The target controller validates the request against its conflict matrix and current restrictions. 4. The controller returns <code>ACK</code> to confirm that the request has been accepted either for execution or for safe deferral; a deferred request must be revalidated before activation. 5. The controller completes the current protected interval and requires clearances. 6. The requested through-movement receives green for the accepted duration. 7. The override expires automatically or is cancelled by the operator. 8. The controller safely terminates the override movement through the required clearances. 9. The controller restores its previous normal mode and reports the result to Central. 10. Central displays completion of the override, ending the use case.
Alternative Flows	3.1 Pedestrian clearance is in progress: 3.1.1. The controller does not truncate the pedestrian sequence. 3.1.2. If the request cannot be safely deferred, the controller ends with <code>NACK</code> . 3.1.3. Otherwise, the controller immediately returns <code>ACK</code> for the retained request, meeting the Central command-response deadline, and waits for clearance to be completed. 3.1.4. Once clearance completes, the controller revalidates the retained request. 3.1.5. If the request remains valid, continue at Step 5 . Otherwise, the retained request is discarded, and the use case ends without applying it. 3.2 Railway pre-emption conflicts with the requested movement: 3.2.1. The controller rejects the request with <code>NACK</code> . 3.2.2. The movement toward the crossing remains red. 3.2.3. Central displays the rejection, and the use case ends. 3.3 Requested duration is invalid or unbounded: 3.3.1. The controller rejects the request or requires a valid bounded duration. 3.3.2. No override is applied. 3.3.3. The use case ends unless the operator submits a corrected request. 6.1 A higher-priority fault occurs during the override: 6.1.1. The controller terminates the override through the applicable safe transition. 6.1.2. It enters the required local fault response. 6.1.3. Central is notified, and the use case ends in the safe fault state.
Post-conditions	1. The override has been completed or been rejected. 2. The controller has returned to its prior normal mode unless a higher-priority state remains active. 3. No local safety invariant has been bypassed.
Business Rules	BR-1: An override may not bypass yellow or all-red clearance (TL-01). BR-2: A movement toward a non-open crossing must receive no green (CC-02). BR-3: An unsafe override must be rejected with <code>NACK</code> (PA-09). BR-4: Central may issue a high-level override request, but the target local controller retains authority to validate the request and actuate its own signal outputs ([8]; PA-09). BR-5: An override must be bounded and auto-expiring (PA-11). BR-6: A safely retained request awaiting pedestrian clearance must receive <code>ACK_ACCEPTED_PENDING</code> within 1 s; only its activation is deferred, and the request must be revalidated before activation (PA-12). BR-7: A renewal request is independently validated and may be rejected with <code>NACK</code> without affecting the existing expiry time (PA-11).

3.2.9 UC-09 - Monitor Network Status

Table 18. Use Case 09 Specification

Field	Description
Name	UC-09 – Monitor Network Status
Goal	Provide the Central Control Room Operator with an accurate network-wide view of controller states and faults.
Summary	Central presents the latest reported state of the nine local controllers and identifies missing, stale, or faulted locations. Monitoring does not grant Central direct hardware-actuation authority.
Primary Actors	Central Control Room Operator
Secondary Actors	None
Pre-conditions	1. Central is operational. 2. At least one local controller is registered for monitoring. 3. The Central display is available to the operator.
Triggers	The operator opens or refreshes the network-status display, or the system receives a state transition, heartbeat timeout, or fault report.
Main Flow	1. The operator opens the network-status display. 2. Central receives current state and health information from the available local controllers. 3. Central associates each report with its physical intersection or railway crossing. 4. The display presents each available controller's mode, signal or crossing state, sensor status, and active faults. 5. Central checks the freshness of each controller using the heartbeat policy. 6. Central identifies any missing, stale, or faulted locations. 7. The operator reviews the current network state and active fault indications. 8. Central continues updating the display as new reports arrive, completing the requested monitoring view.
Alternative Flows	2.1 A previously unavailable controller reconnects: 2.1.1. Central waits for the controller's complete current state report. 2.1.2. Central replaces the stale view only after receiving that report. 2.1.3. Return to Step 3. 5.1 A controller becomes unreachable: 5.1.1. Central marks that controller as stale or unavailable. 5.1.2. The last report is not presented as current state. 5.1.3. Continue at Step 6. 6.1 A railway fault report arrives: 6.1.1. Central displays the fault with the applicable severity. 6.1.2. The railway controller independently maintains its local safe state. 6.1.3. Continue at Step 7.
Post-conditions	1. The operator can distinguish current, stale, unavailable, and faulted controller states. 2. Newly reported faults are visible to the operator. 3. Monitoring has not directly altered any local actuator.
Business Rules	BR-1: Central monitors controller status but does not directly control intersection lights. The three railway crossings are independently managed by RL1–RL3 (RC-02). [8] BR-2: Heartbeat loss is identified after three consecutive missed heartbeats (PA-07). BR-3: A reconnected controller supplies its current state before accepting new commands (PA-08) BR-4: Railway fault reporting must not delay the local STOP response (RC-10).

3.2.10 UC-10 - Continue Local Operation

Table 19. Use Case 10 Specification

Field	Description
Name	UC-10 – Continue Local Operation

Goal	Maintain safe local operation while Central communication is unavailable and safely resynchronize after reconnection.
Summary	A local controller detects loss of its Central link and continues autonomously. It retains its last validated parameters, uses its local clock for mode selection, and sends its actual state to Central after reconnection.
Primary Actors	Communication network
Secondary Actors	Central Control Room Operator
Pre-conditions	1. The local controller operates normally. 2. It has a previously validated timing configuration. 3. It normally exchanges heartbeats with Central.
Triggers	Three consecutive 1-second Central heartbeats are missed.
Main Flow	1. The local controller detects the heartbeat timeout. 2. The controller declares its Central link unavailable and enters <code>DEGRADED_LOCAL</code>. 3. Central coordination and override availability are disabled for that controller. 4. The controller retains its last validated timing parameters. 5. Its local clock continues selecting <code>PEAK_FIXED</code> or <code>OFF_PEAK_SENSOR</code> as applicable. 6. The affected local controller continues its applicable local control and safety functions without Central dependency. 7. The communication link becomes available again. 8. The local controller sends its complete current state to Central. 9. After the state exchange completes, the controller accepts new Central requests and normal coordination resumes. 10. The controller exits <code>DEGRADED_LOCAL</code>, completing the recovery.
Alternative Flows	6.1 Railway events occur while disconnected: 6.1.1. The relevant local railway controller executes the complete railway-protection sequence. 6.1.2. Adjacent intersections apply the required railway restrictions. 6.1.3. No action waits for Central. 6.1.4. Return to Step 7 when communication becomes available. 7.1 Link remains unavailable: 7.1.1. The controller continues autonomous operation using its last validated parameters. 7.1.2. Its local clock continues normal Peak/Off-Peak selection. 7.1.3. The use case ends in safe <code>DEGRADED_LOCAL</code> operation. 8.1 Central holds stale state after reconnection: 8.1.1. The local controller rejects or defers new Central requests. 8.1.2. The controller completes transmission of its current state. 8.1.3. Return to Step 9 after successful state synchronization.
Post-conditions	1. The controller is continuing safe autonomous operation; or 2. It has transferred its actual state to Central and resumed validated coordination. 3. Local operation has not frozen or entered an unnecessary all-red state solely because Central was unavailable.
Business Rules	BR-1: Heartbeats occur every second, and three consecutive misses declare the Central link unavailable (PA-07). BR-2: The controller retains its last validated parameters while its local clock continues selecting the applicable mode (PA-07). BR-3: On reconnection, the controller sends its current state before accepting new Central commands (PA-08). BR-4: Loss of coordination must degrade to standalone operation rather than blocking the controller (TC-04).

4. System Behavior and State Models

The section defines the system's dynamic behavior under normal, coordinated, preempted, and faulty conditions. The state charts specify the valid states and transitions of the intersection, pedestrian, railway-crossing, and communication controllers, while the sequence diagrams show how distributed processes exchange messages to realize the principle use-case scenarios

4.1 Intersection State Charts

The following state charts define the traffic-light, pedestrian, railway-crossing, and communication behavior of the distributed traffic-control system. Generic charts are used because *L1-L6* share one configuration-driven intersection-controller design and *RL1-RL3* share one railway-controller design (**RC-07**). Together, they cover normal operations, safety intervention, equipment faults, and loss of communication with Central.

Where a single state chart would otherwise be too dense to remain legible on an A4 page, it is divided into lettered sub-diagrams (*SC-01A/B/C*, *SC-03A/B*, and *SC-04A/B*). These sub-diagrams connect through named UML entry and exit points or reference the detailed behavior of a composite state. Full-resolution versions of all state-chart diagrams are available through the [online supplementary folder](#) for clearer viewing.

Notation note: The state charts follow the UML state-machine conventions described by draw.io. Transition labels use the format **trigger** [**guard condition**] / **action**. Therefore, / separates the event or condition that enables a transition from the action performed when that transition occurs. [10]

4.1.1 SC-01A - Vehicle-Signal Mode Selection Overview

Figure 6 shows how a generic intersection controller chooses between *PEAK_FIXED* and *OFF_PEAK_SENSOR* without repeating the phase-by-phase behavior of either mode. A pending mode change, whether selected by the local clock or requested through a Central *SET_MODE* command, is applied only after the current phase and its required clearances have completed (**TL-04**). *SC-01B* defines the *PEAK_FIXED* cycle, while *SC-01C* defines the *OFF_PEAK_SENSOR* cycle. This chart supports UC-01 and UC-07.

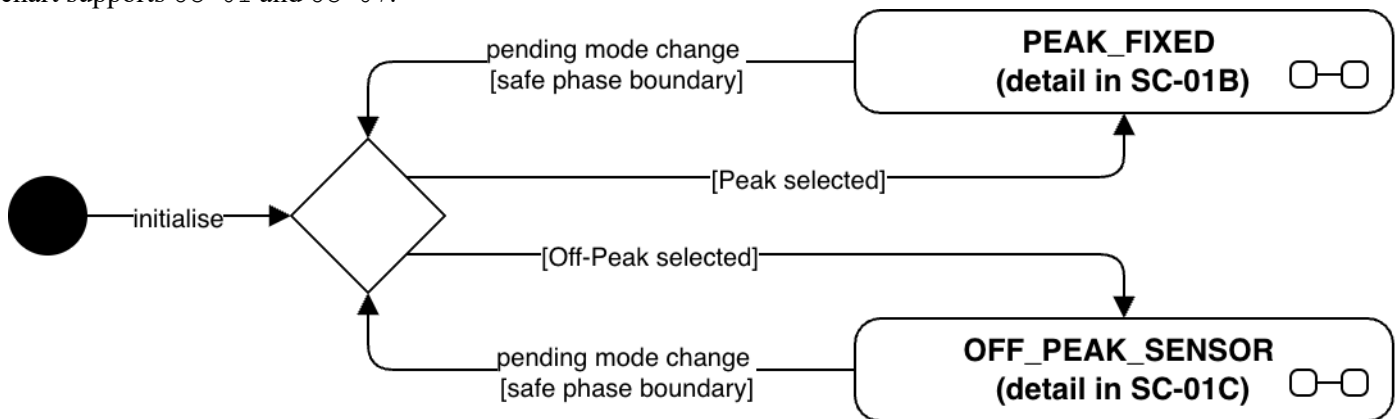


Figure 6. SC-01A State Diagram

4.1.2 SC-01B - *PEAK_FIXED* Phase Detail

Figure 7 details the fixed 90-second two-phase cycle used while *PEAK_FIXED* is active. Ordinary approach sensors are monitored for status and fault reporting but do not alter the configured phase durations (**TL-01**, **TL-02**).

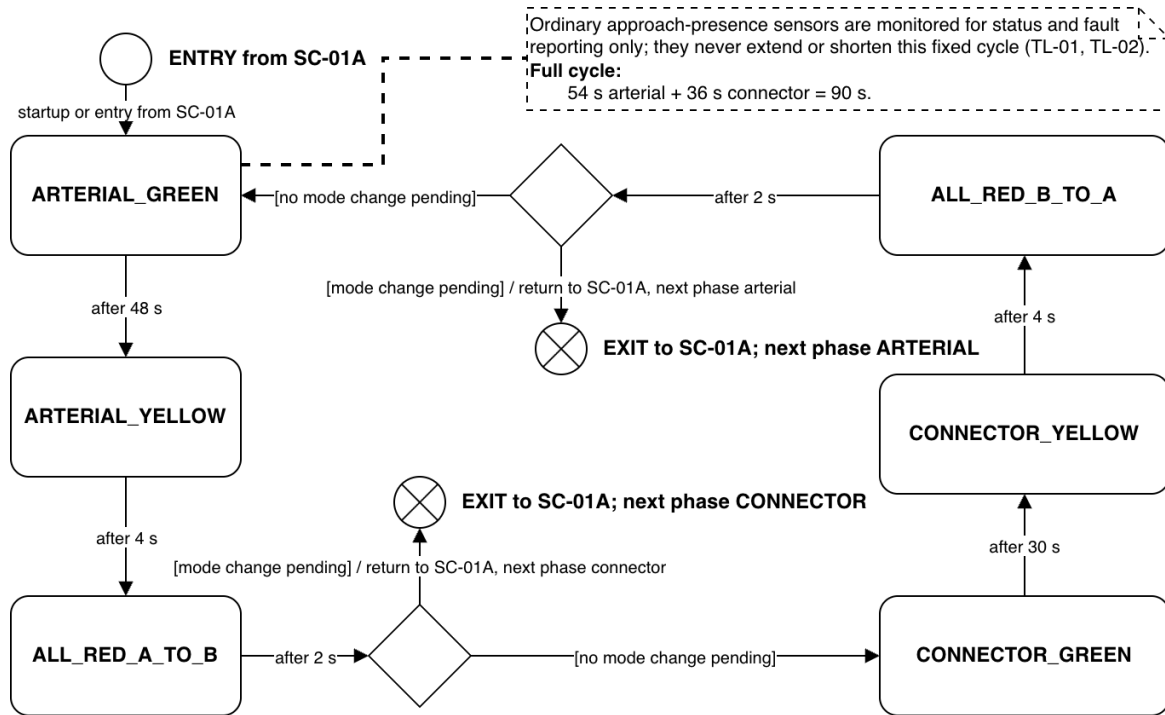


Figure 7. SC-01B State Diagram

4.1.3 SC-01C - OFF_PEAK_SENSOR Phase Detail

Figure 8 illustrates the demand-responsive cycle used while `OFF_PEAK_SENSOR` is active. Each green interval is bounded between 8 and 40 seconds and extends only while demand persists; the arterial cap ensures that a pending connector request is served after no more than one further arterial service opportunity (`TL-03`, `DP-04`, `DP-06`).

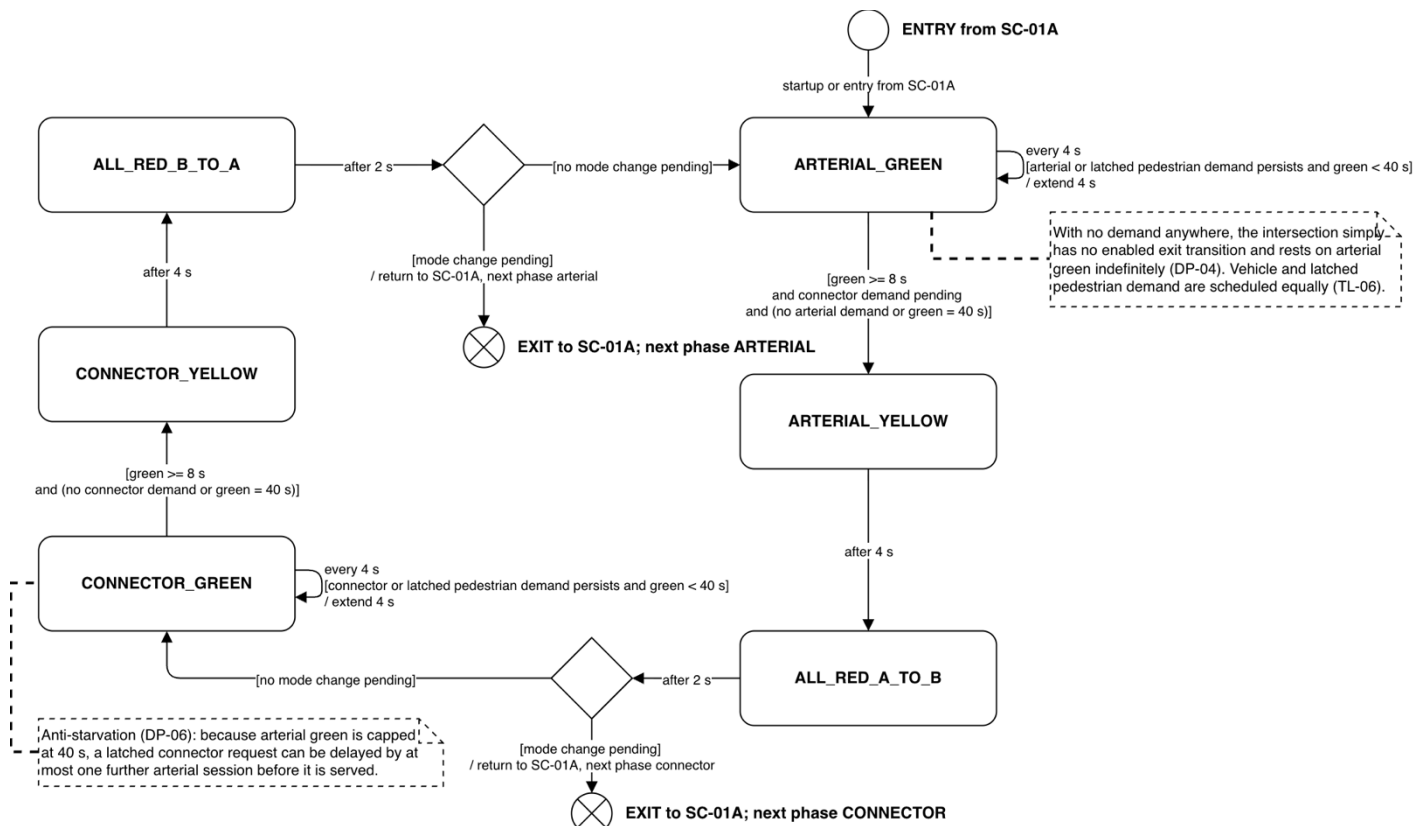
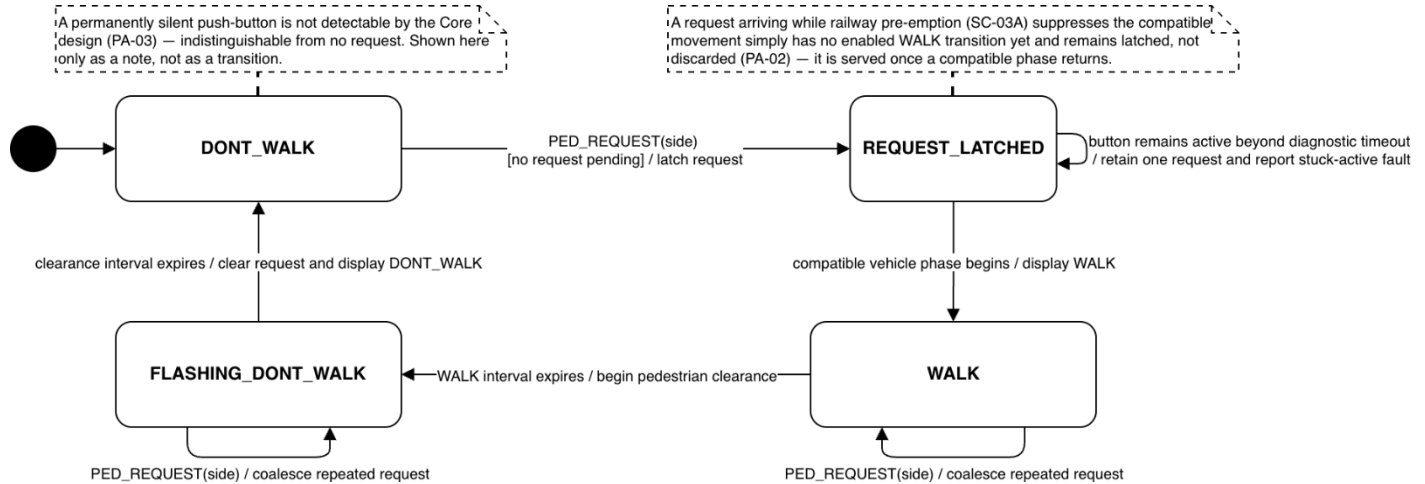


Figure 8. SC-01C State Diagram

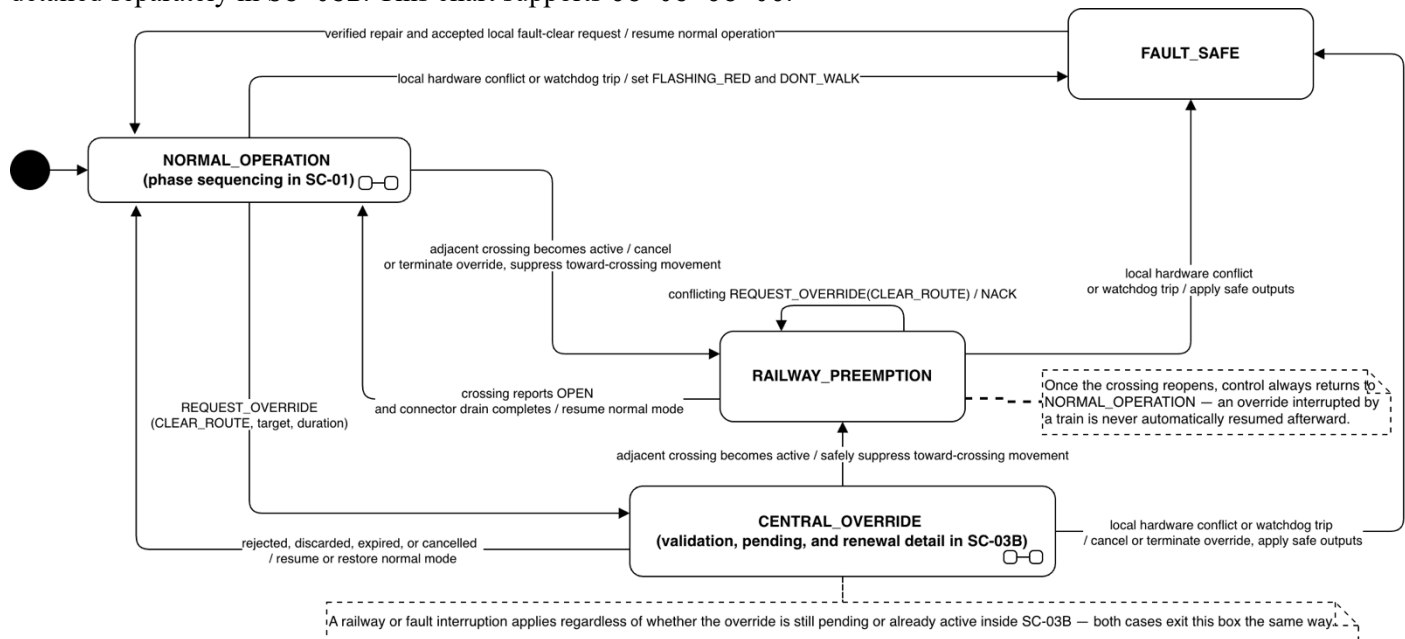
4.1.4 SC-02 - Generic Pedestrian-Signal State Chart

Figure 9 shows the lifecycle of a pedestrian request at one crossing side. A button press is latched until a compatible vehicle phase becomes available, after which the complete `WALK` → `FLASHING_DONT_WALK` → `DONT_WALK` sequence runs without unsafe interruption (**TL-05**, **TL-06**, **PA-02**, **PA-03**). This chart supports UC-02 and the pedestrian-safety constraint in UC-08.



4.1.5 SC-03A - Intersection Supervisory Authority Overview

Figure 10 presents the supervisory authority conditions that control an intersection above the normal phase sequencer in SC-01. Local fault protection has the highest authority, followed by railway pre-emption, a validated Central override, and normal Peak or Off-Peak operation. Every request is checked locally, and no Central command may bypass railway or pedestrian safety constraints (**PA-09**, **PA-10**). The validation, queueing, activation, and renewal of an override are detailed separately in SC-03B. This chart supports UC-05–UC-08.



4.1.6 SC-03B - Clear-Route Override Validation, Pending, and Renewal Detail

Figure 11 details the `CENTRAL_OVERRIDE` state introduced in SC-03A. It shows local validation, safe deferral behind an active pedestrian clearance, bounded activation, renewal, cancellation, and automatic expiry. A safely deferred request

is acknowledged promptly, but it is activated only after pedestrian clearance completes and the request remains valid (PA-09, PA-11).

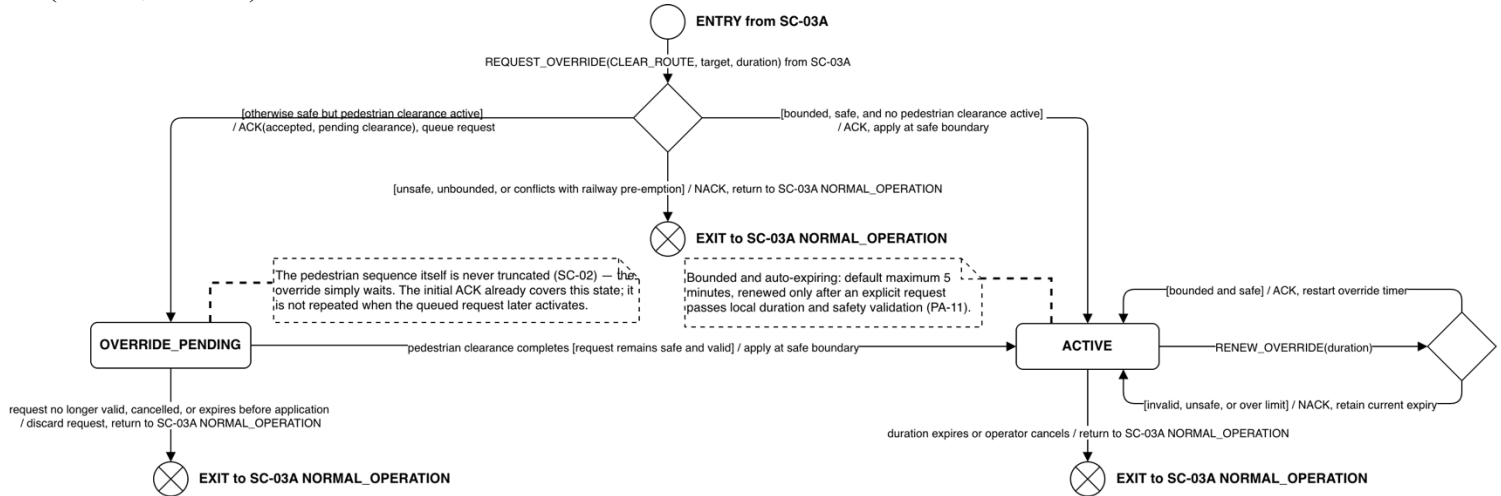


Figure 11. SC-03B State Diagram

4.1.7 SC-04A – Railway Crossing Approach and Closure

Figure 12 shows the first part of the railway-crossing lifecycle shared by RL1–RL3. It covers train-approach detection, the flash-only warning interval, commanded gate closure, and sensor confirmed closure before any train signal may display PROCEED. The lifecycle continues in SC-04B after the gates are confirmed CLOSED (RC-03, RC-06, RC-11).

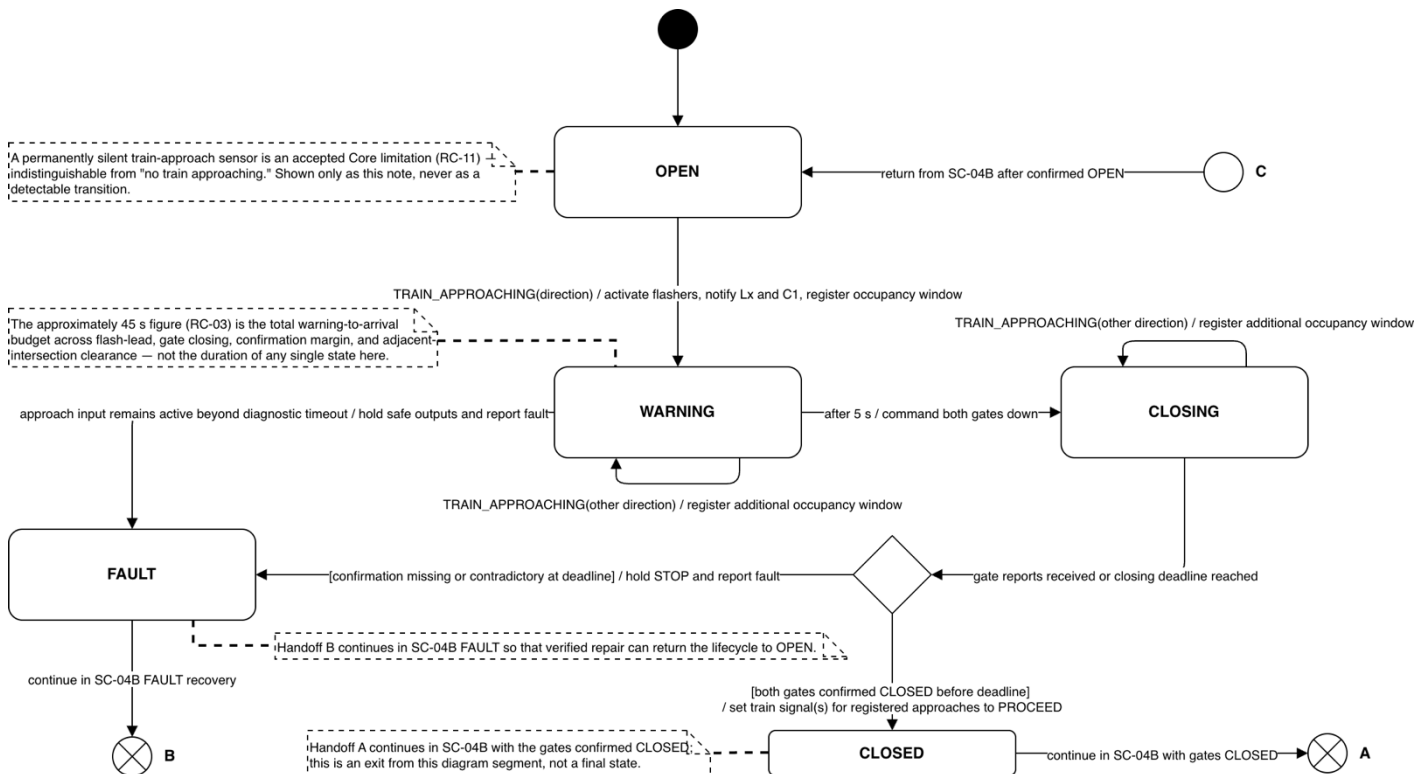


Figure 12. SC-04A State Diagram

4.1.8 SC-04B - Railway Crossing Occupancy and Reopening

Figure 13 continues the second part of the railway-crossing lifecycle, entered from SC-04A after gate closure is confirmed. The gates remain closed while any registered occupancy window is active, including windows created by additional trains from either direction, and all train signals return to STOP before the gates begin opening (RC-04, RC-05).

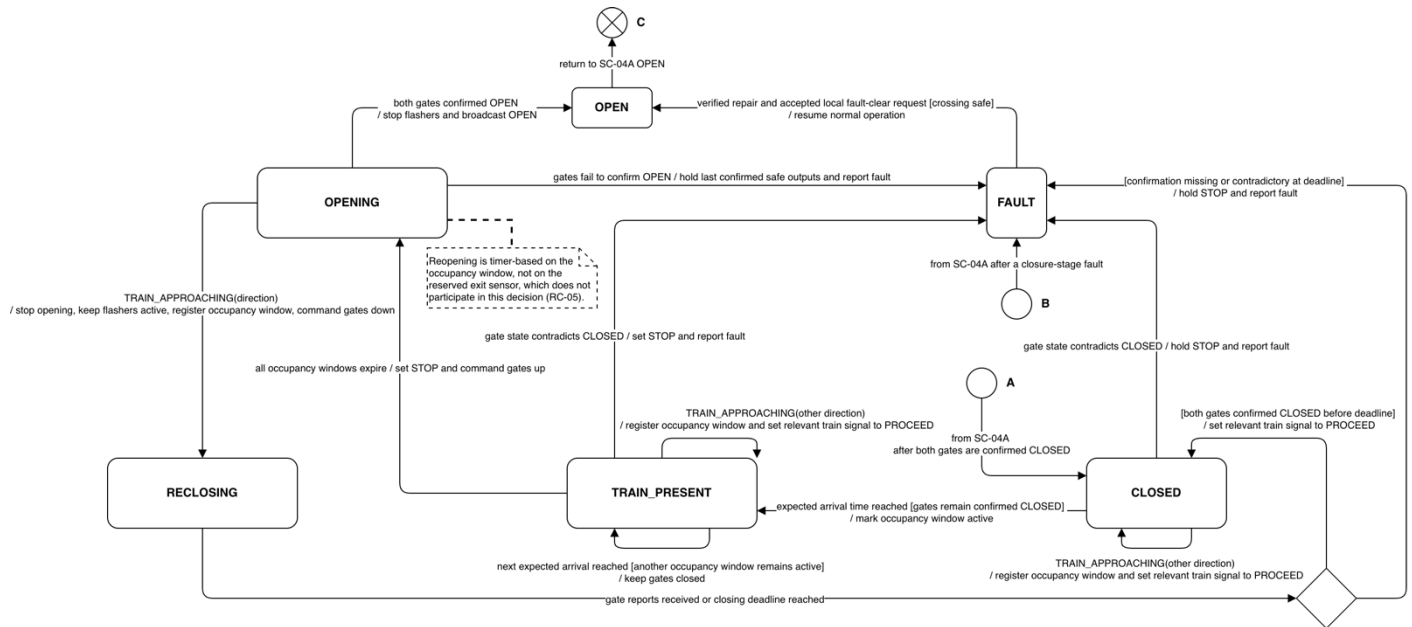


Figure 13. SC-04B State Diagram

4.1.9 SC-05 - Central Connectivity and Local Autonomy

Figure 14 shows how a local intersection or railway controller responds to changes in its connection with Central. After 3 missed heartbeats, the controller continues autonomously using its last validated parameters. After communication is restored, it reports its complete current state before accepting new Central commands (PA-07, PA-08). This chart supports UC-09 and UC-10.

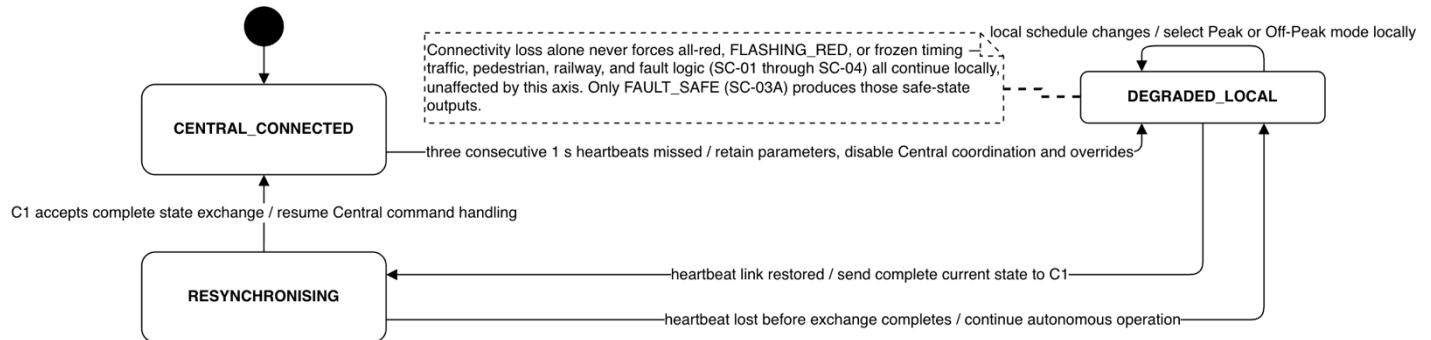


Figure 14. SC-05 State Diagram

4.2 Behavioral Specifications

The following sequence diagrams show the chronological interactions that realize the ten use cases defined in Section 3.2. Eight diagrams provide complete coverage because UC-03 and UC-07 share the same configuration distribution interaction, while UC-09 and UC-10 form one continuous monitoring, disconnection, and reconnection sequence. Internal controller states and transitions are defined separately in Section 4.1. Full-resolution versions of all sequence diagrams are available through the [online supplementary folder](#) for clearer viewing.

4.2.1 SD-01 - Serve Off-Peak Vehicle Demand

Figure 15 shows how a generic intersection controller serves vehicle demand in OFF_PEAK_SENSOR. It includes demand latching, mandatory clearance intervals, bounded green extension, anti-starvation, and the arterial rest condition (TL-01, TL-03, DP-04, DP-06, PA-04). It realizes UC-01.

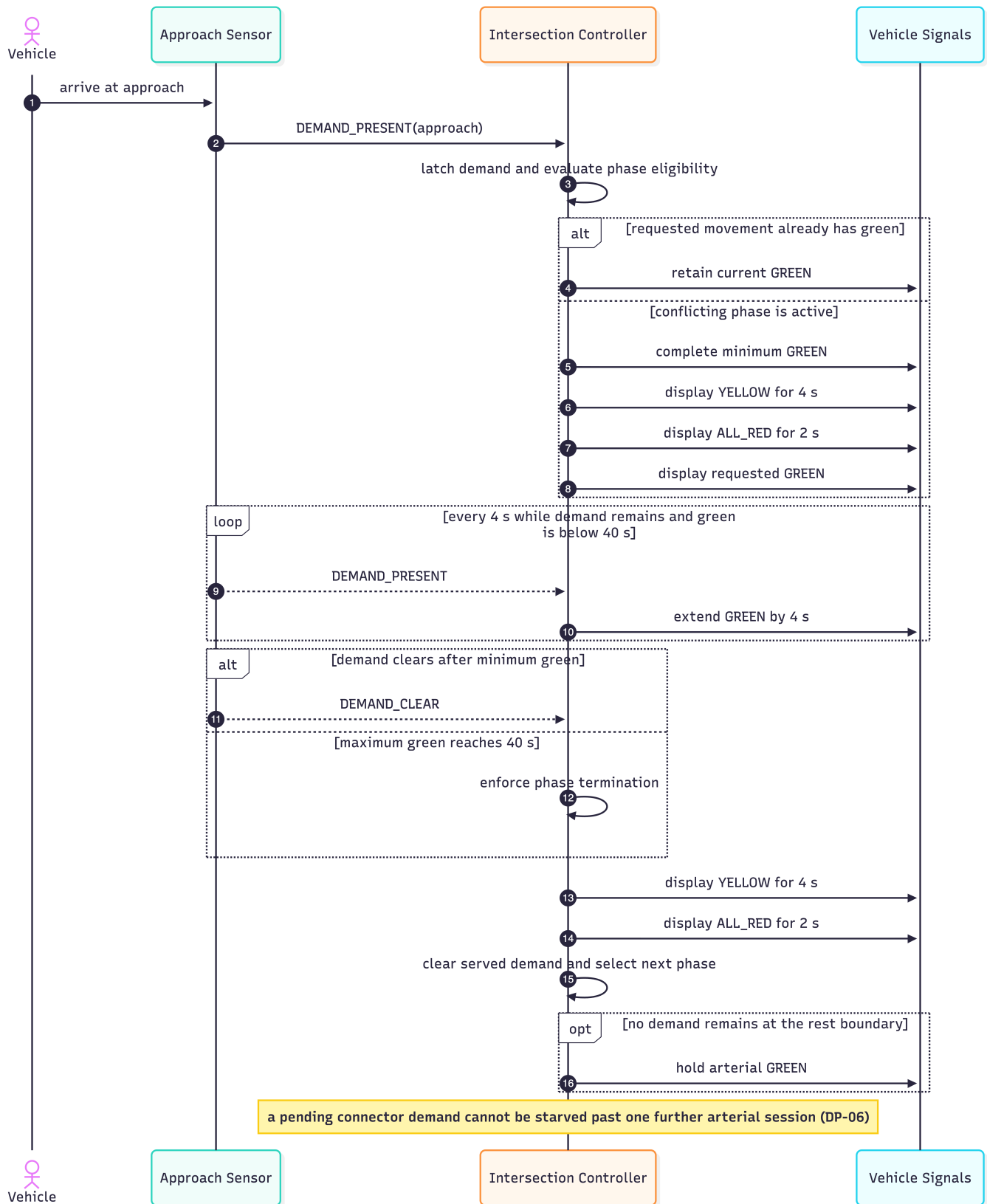


Figure 15. SD-01 Sequence Diagram

4.2.2 SD-02 - Serve a Pedestrian Crossing Request

Figure 16 presents the complete lifecycle of one pedestrian request. The request is latched until a compatible phase is available, repeated presses are coalesced, and the full pedestrian clearance completes before a conflicting vehicle movement may begin (TL-05, TL-06, PA-02, PA-03). It realizes UC-02 and the pedestrian-safety constraint used by UC-08.

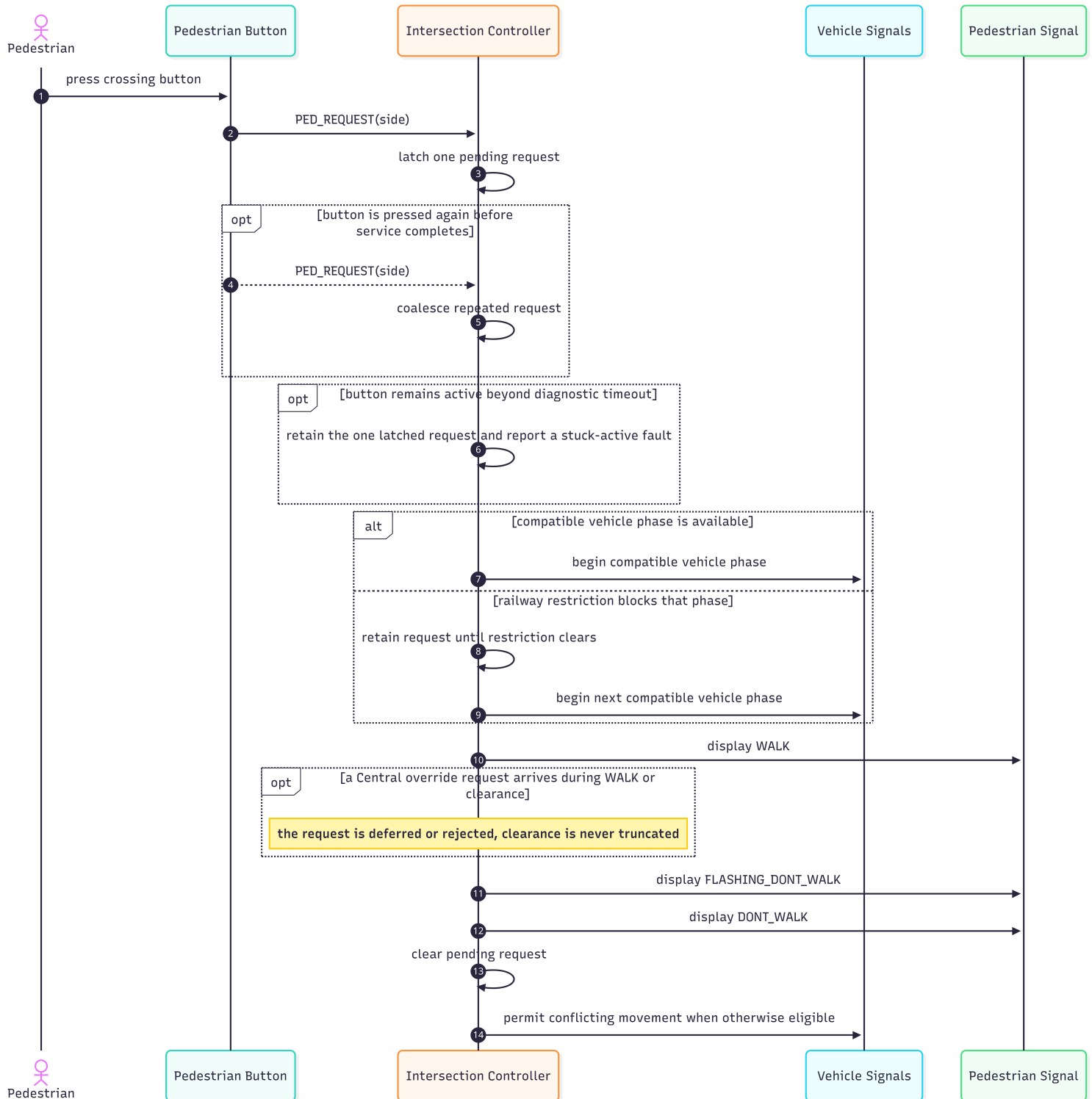


Figure 16. SD-02 Sequence Diagram

4.2.3 SD-03 - Configure and Apply an Arterial Coordination Profile

Figure 17 shows Central distributing a validated timing profile to an arterial controller chain. Each local controller independently validates its assigned offset, returns ACK or NACK, applies an accepted profile only at a safe phase boundary, and reports its resulting state (TC-01-TC-05, TL-04, PA-09). The same interaction supports a SET_MODE request under UC-07. It realizes UC-03 and UC-07.

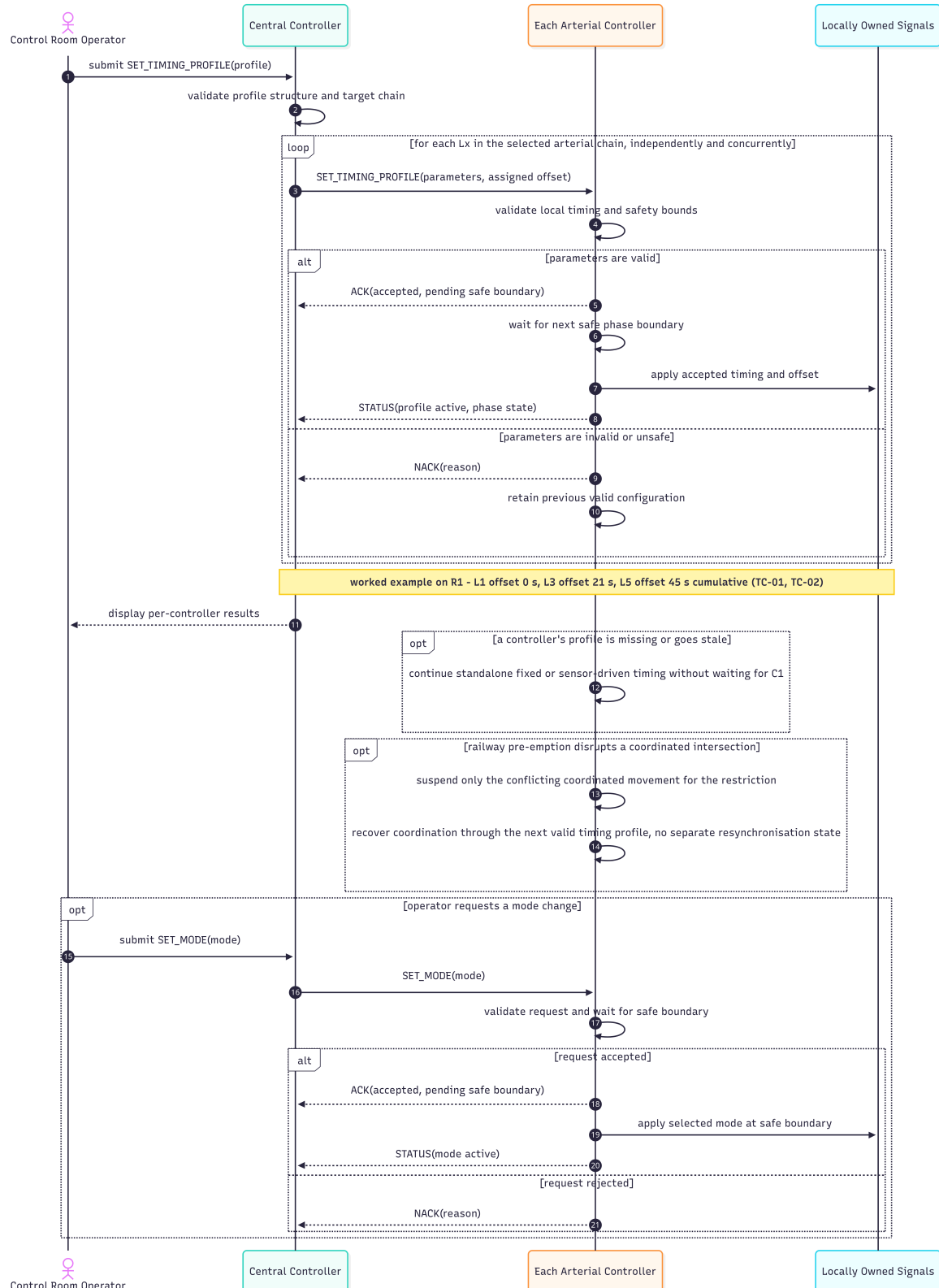


Figure 17. SD-03 Sequence Diagram

4.2.4 SD-04 - Protect a Railway Crossing for an Approaching Train

Figure 18 presents the complete protection cycle at one railway crossing. The railway controller activates warnings locally, informs the adjacent intersections, requires physical gate-closed confirmation before displaying **PROCEED**, tracks overlapping train-occupancy windows, and confirms that the gates are open before ending the warning (**RC-01-RC-06**). It realizes UC-04.

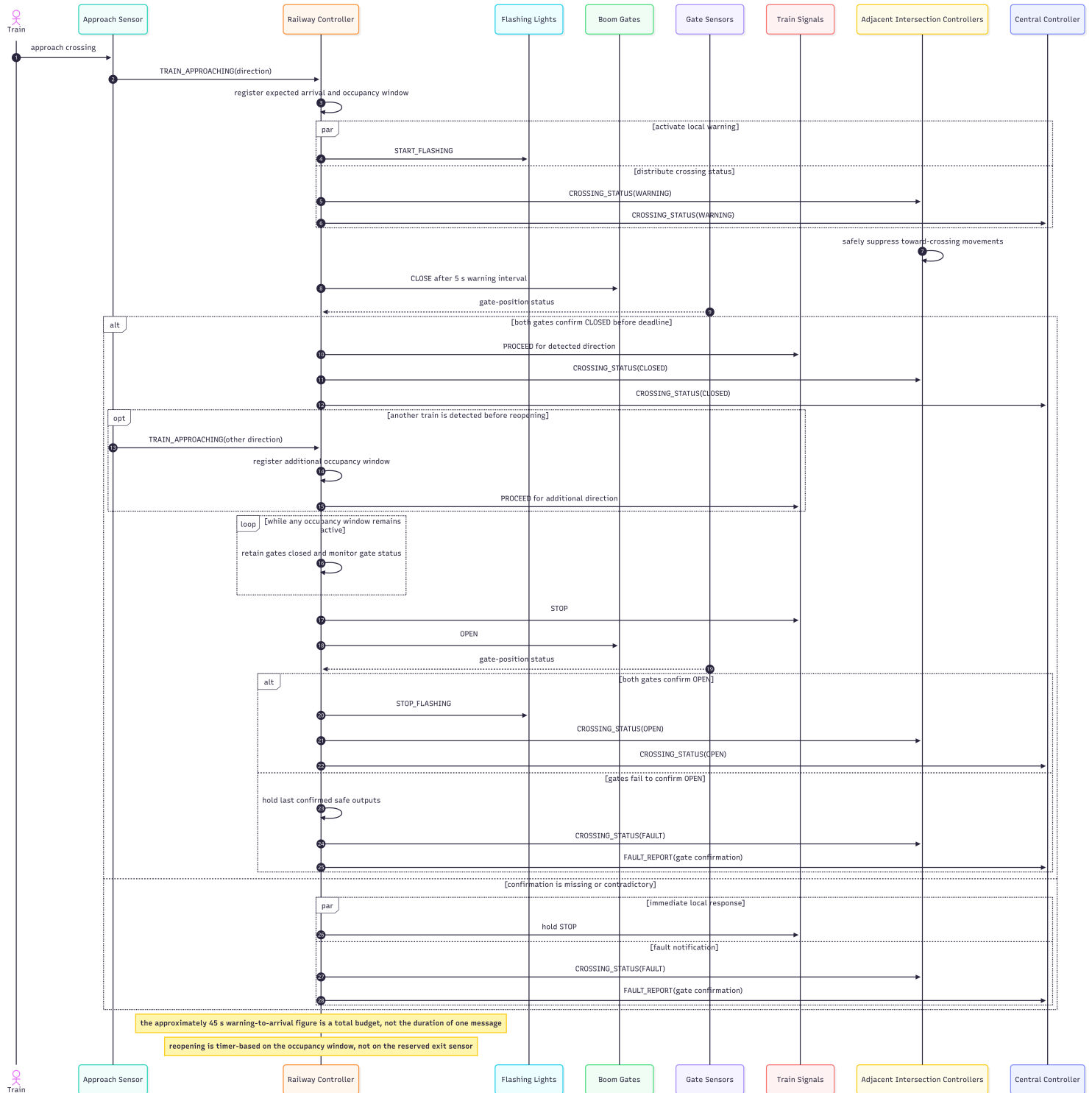


Figure 18. SD-04 Sequence Diagram

4.2.5 SD-05 - Suppress and Drain Road Traffic Around a Railway Closure

Figure 19 shows the behavior independently executed by each of the two intersection controllers adjacent to a railway crossing. Each controller safely suppresses its toward-crossing movement while the crossing is unavailable, then uses its binary queue detector to run a drain phase in 4-second increments, capped at 60 seconds (CC-01–CC-03, TL-01). It realizes UC-05.

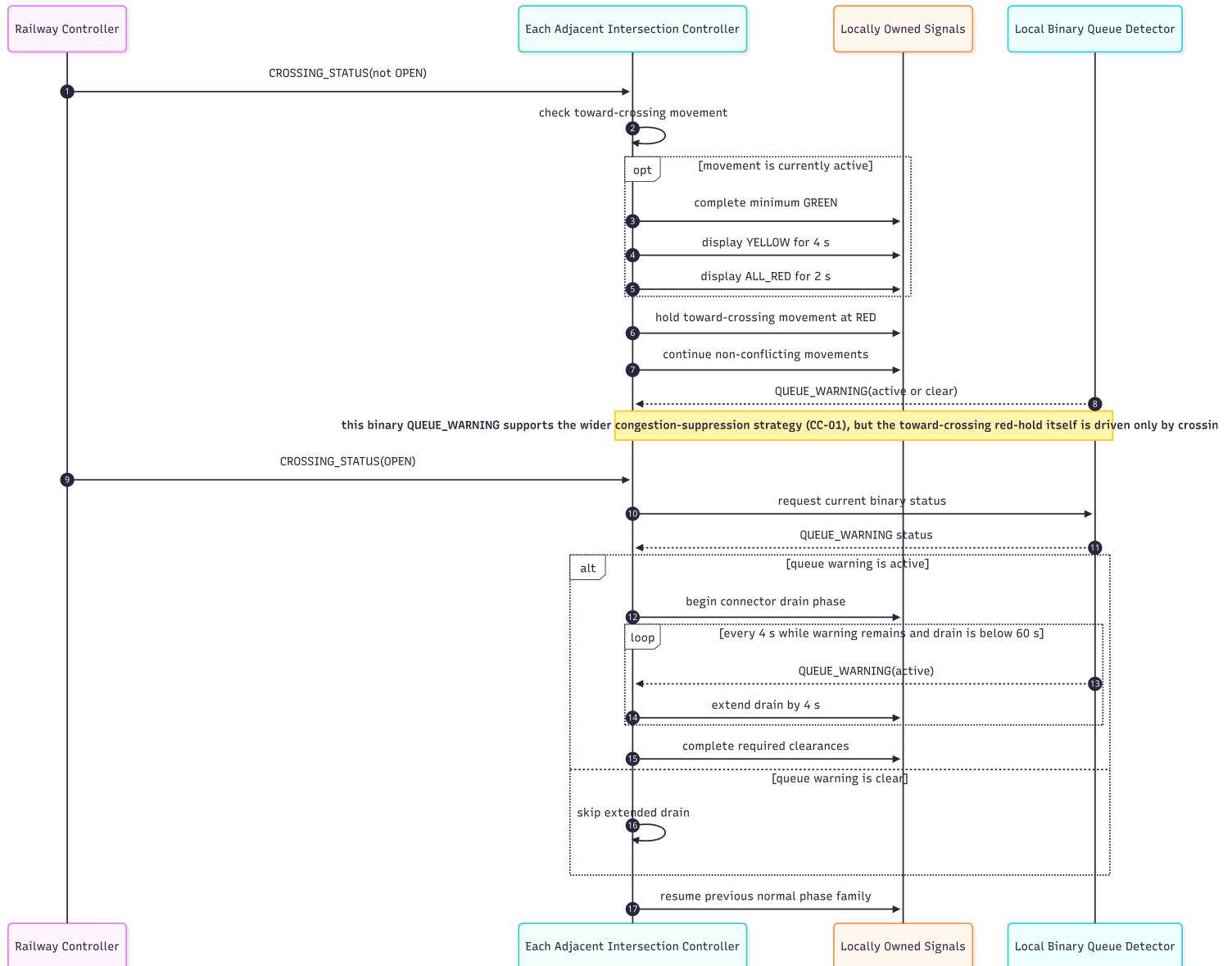


Figure 19. SD-05 Sequence Diagram

4.2.6 SD-06 - Contain and Report a Railway Equipment Fault

Figure 20 shows a railway controller responding to unverifiable gate condition. The train signal is forced to STOP locally while the fault is reported in parallel, ensuring that the safety response never depends on Central communication. Normal operation resumes only after physical repair is verified and a fault-clear request is accepted (RC-06, RC-09, RC-10, PA-09, PA-10). It realizes UC-06.

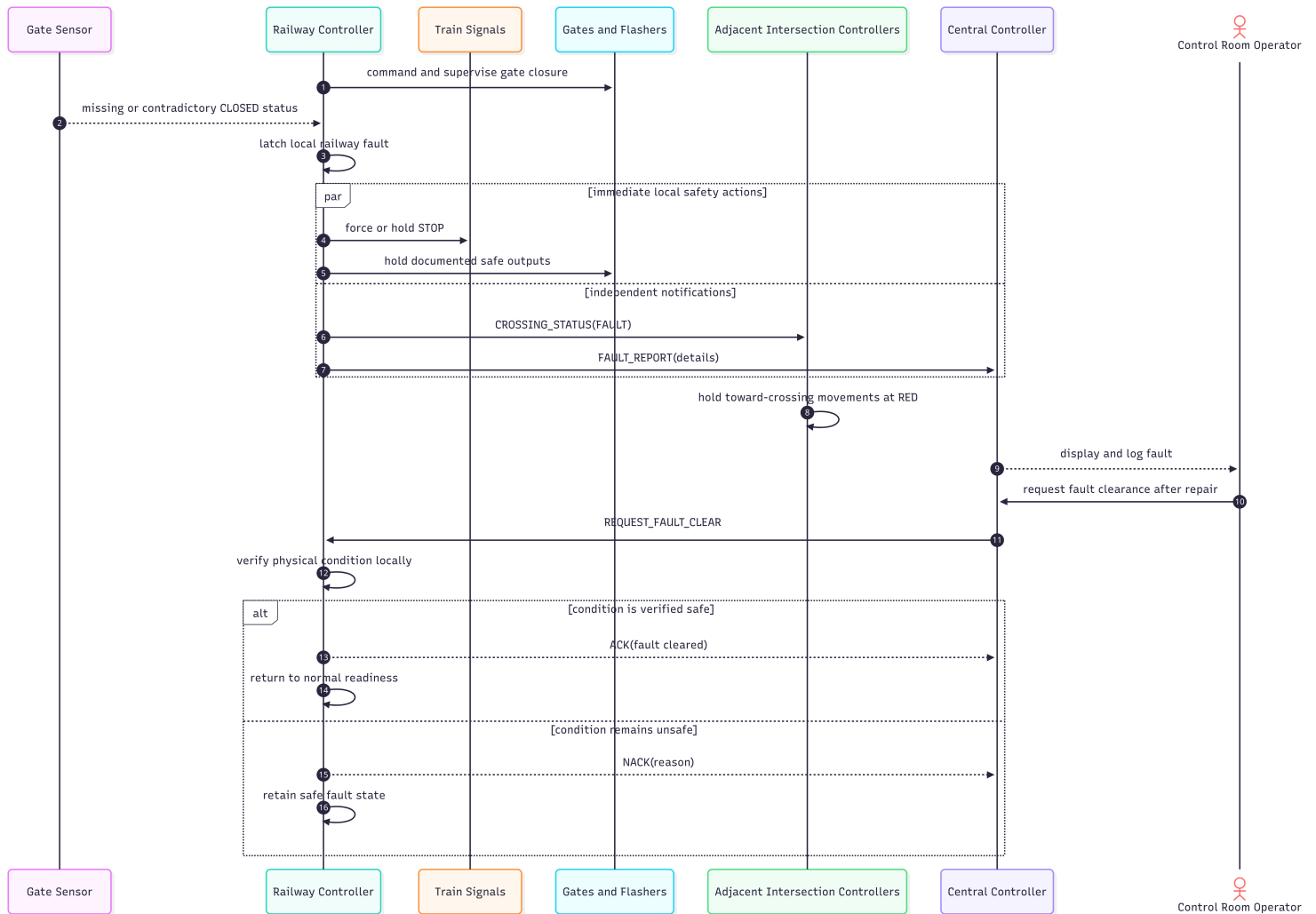


Figure 20. SD-06 Sequence Diagram

4.2.7 SD-07 - Validate and Apply a Bounded Clear-Route Override

Figures 21-24 collectively show how a target intersection controller validates and executes a Central clear-route request while retaining exclusive authority over its signals. A safely deferred request is acknowledged within the defined command-response deadline but activates only after pedestrian clearance completes and local validation still succeeds. Initial requests and renewals remain bounded by local safety checks, and every accepted override ends through the required vehicle-signal clearance sequence (PA-09, PA-11, PA-12, UC-08).

Note: Due to the size and level of detail of the sequence diagram, it is presented across four consecutive figures to preserve readability within the report. A full-resolution version is also available for clearer viewing: [View the full-resolution sequence diagram](#).

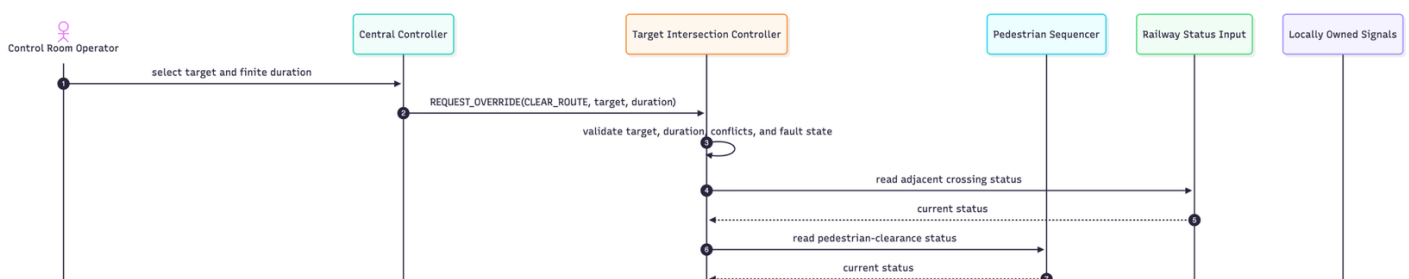


Figure 21. SD-07 Sequence Diagram (1)

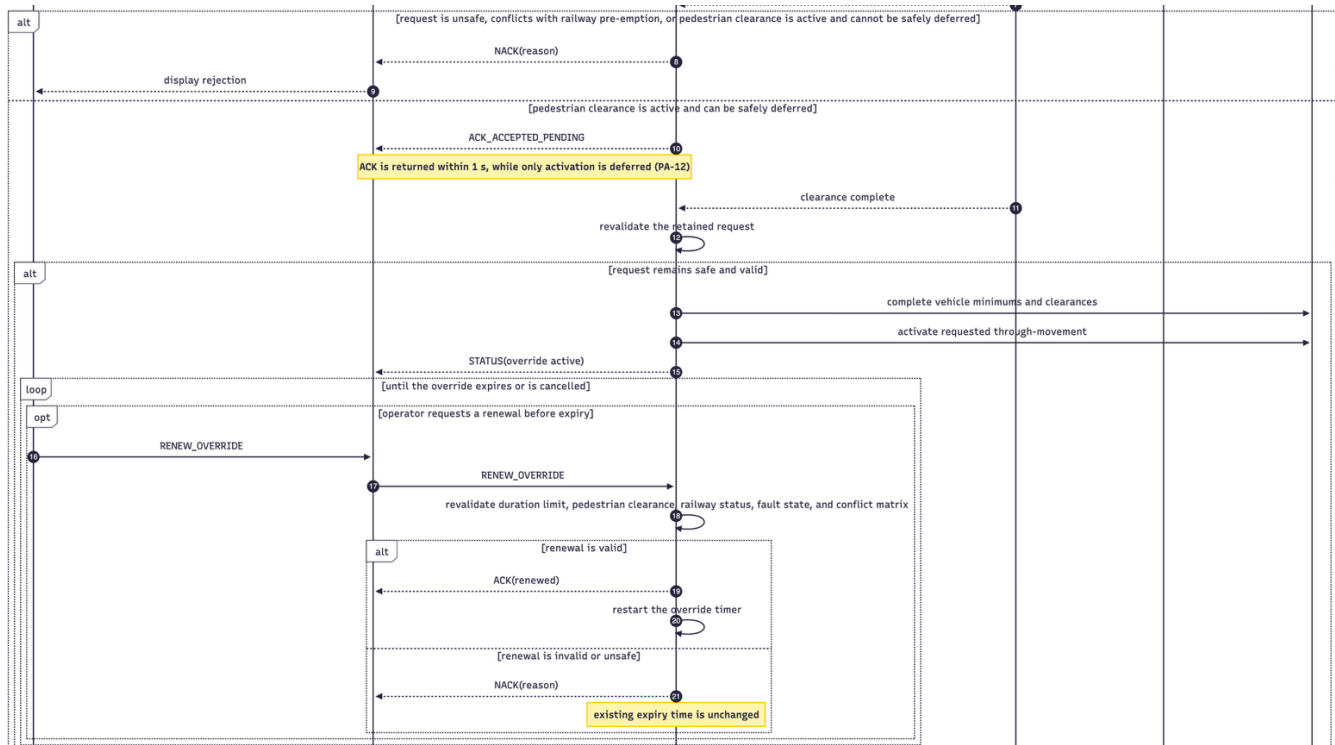


Figure 22. SD-07 Sequence Diagram (2)

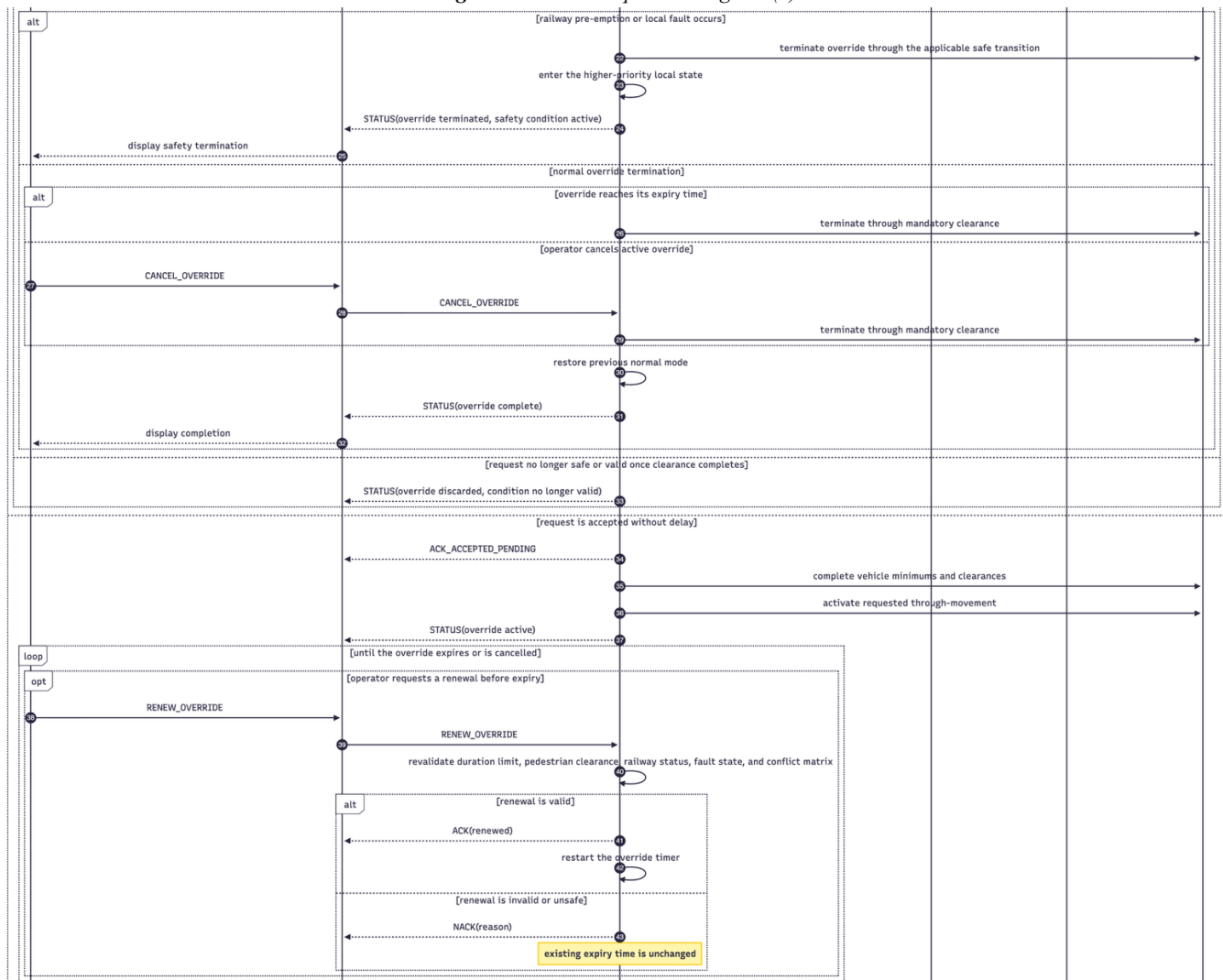


Figure 23. SD-07 Sequence Diagram (3)

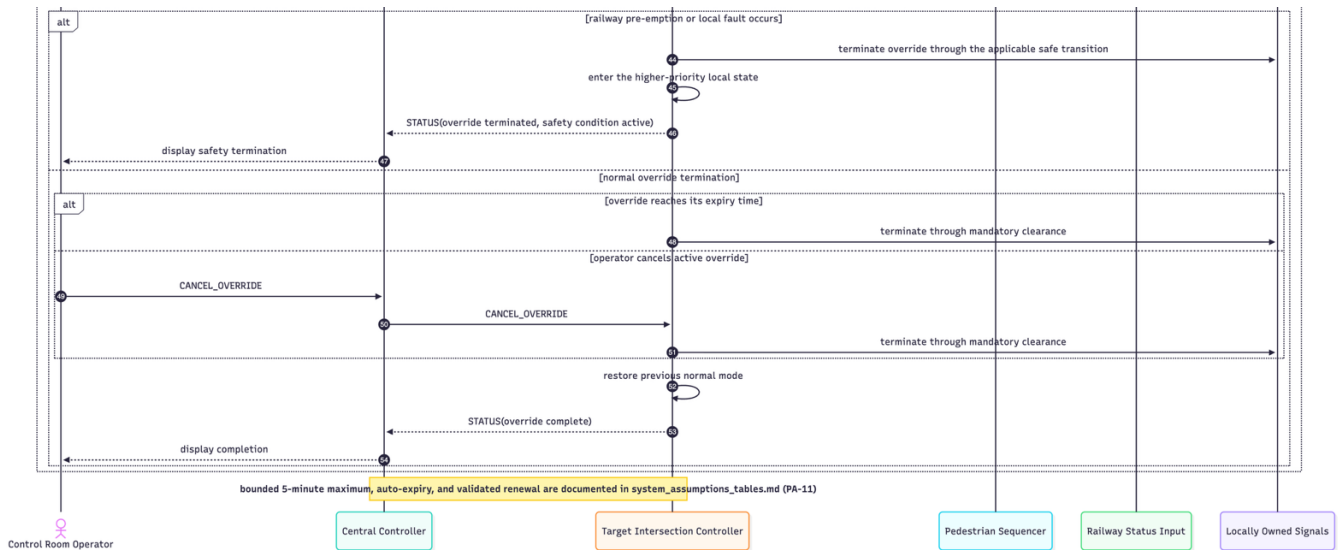


Figure 24. SD-07 Sequence Diagram (4)

4.2.8 SD-08 - Monitor Controllers, Lose Central Link, and Resynchronize

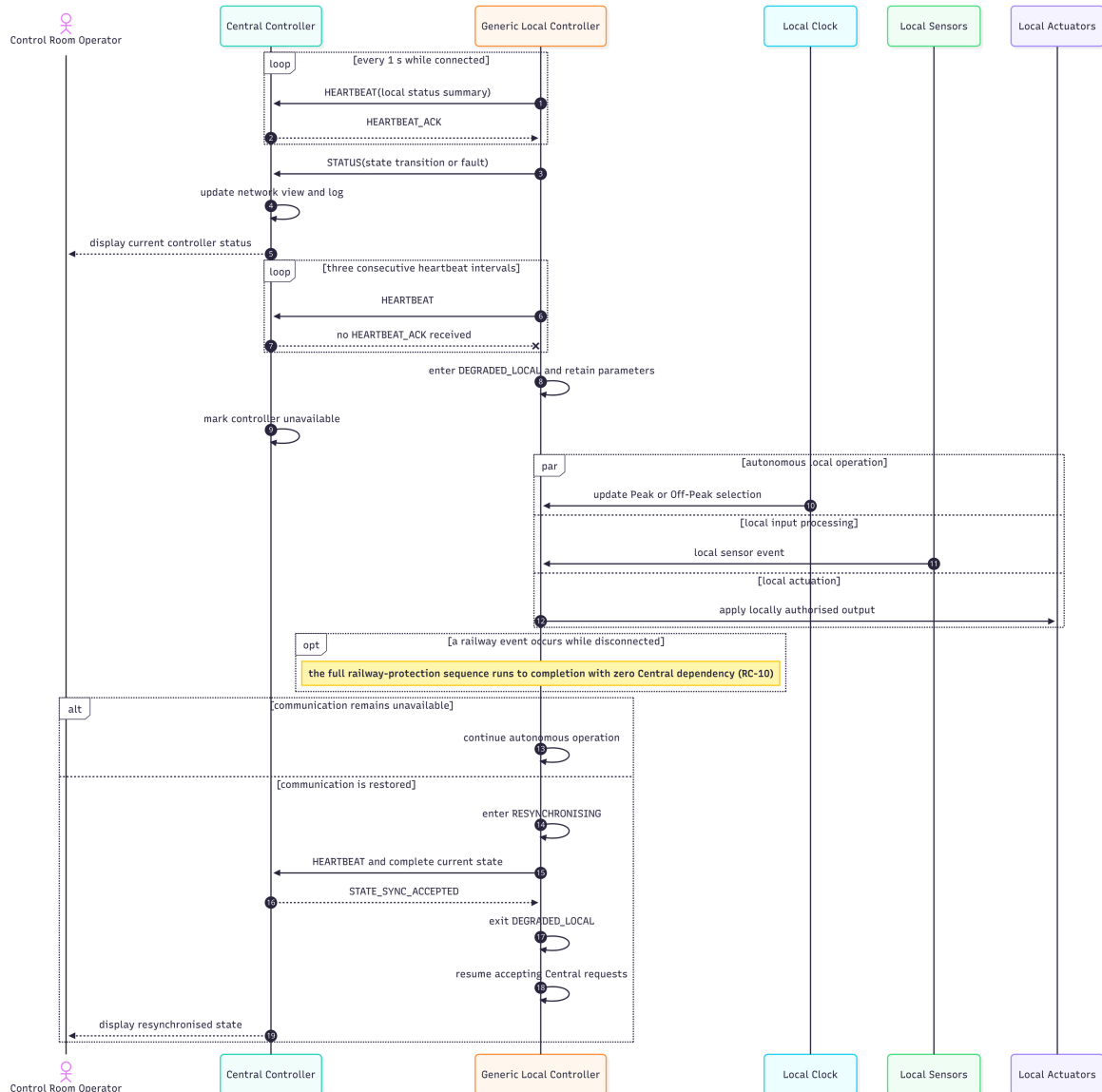


Figure 25. SD-08 Sequence Diagram

Figure 25 shows the combined monitoring, disconnection, and reconnection sequence. Each local controller enters `DEGRADED_LOCAL` after three missed heartbeat acknowledgements, continues its complete local control logic using retained parameters, and sends its current state to Central before accepting new commands after reconnection (**PA-07**, **PA-08**, **TC-04**, **RC-10**). It realizes UC-09 and UC-10

5. Implementation Architecture

The system uses a distributed, locally autonomous architecture comprising **10 logical QNX controller nodes** deployed across two physical computing machines: **one Central Controller (C1)**, **six intersection controllers (L1–L6)**, and **three railway-crossing controllers (RL1–RL3)**. Each local controller retains exclusive authority over its own physical outputs and continues its applicable control and safety functions if Central becomes unavailable.

C1 provides network-wide monitoring, persistent event logging, configurable Peak/Off-Peak mode selection, timing-profile distribution, and bounded override requests. It does not directly actuate traffic lights, boom gates, flashing lights, or train signals. Every Central command is validated locally and receives an ACK or NACK.

Each L_x independently processes vehicle detectors, queue detectors, pedestrian buttons, local timing, traffic-signal sequencing, and railway-crossing status received directly from the relevant RL_x . Each RL_x independently detects approaching trains, controls its boom gates, road-facing flashers, and train signals, and transmits crossing status directly to its two adjacent intersection controllers. This direct safety path does not depend on C1.

5.1 Task Architecture Diagram

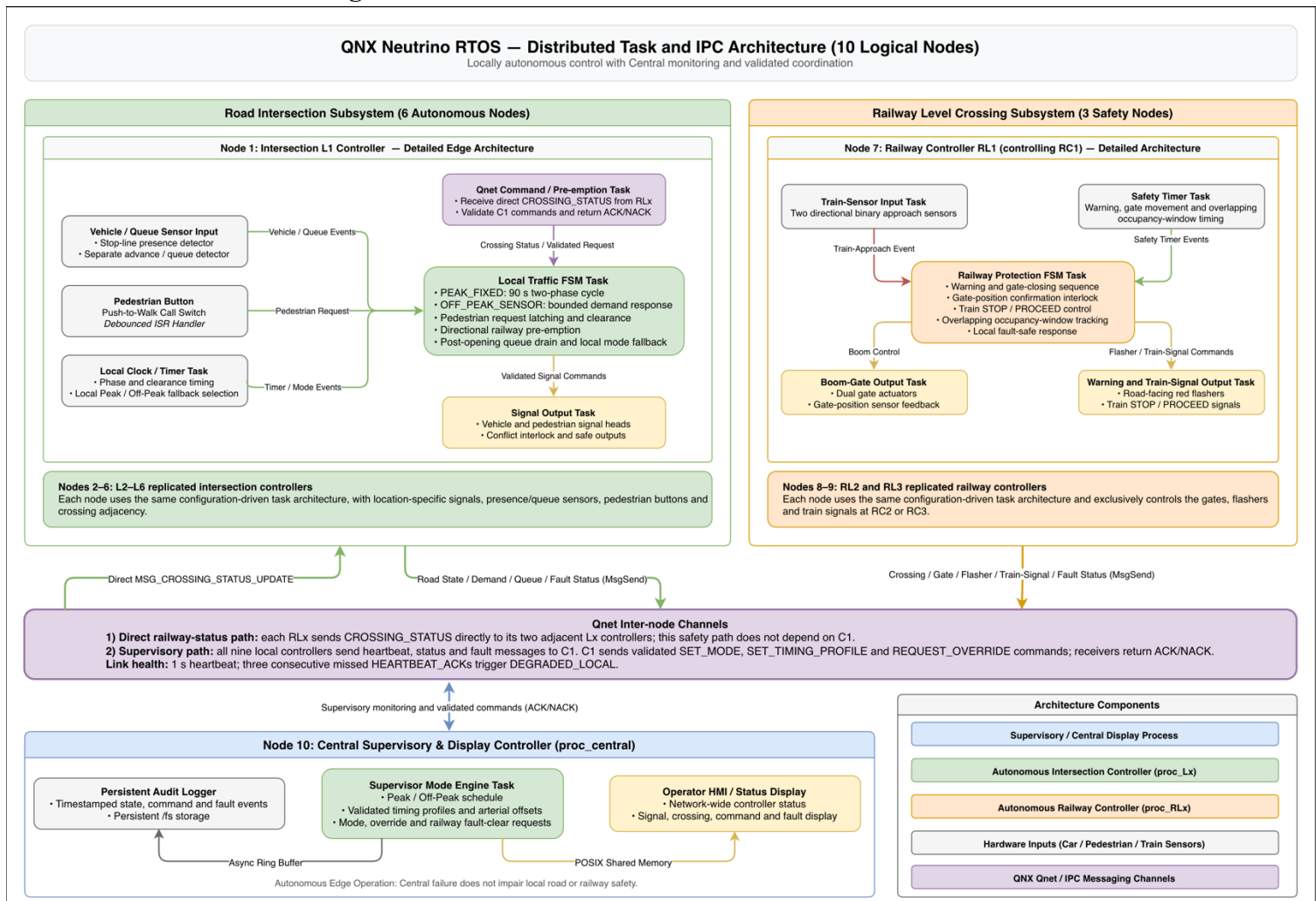


Figure 26. Task Architecture Diagram

Figure 26 presents *the task and IPC architecture of the 10 logical QNX controller nodes*. The detailed L1 and RL1 structures represent configuration-driven controller designs replicated across L2–L6 and RL2–RL3, respectively. The logical nodes are deployed across two physical computing machines without assuming a one-controller-per-computer mapping.

Railway pre-emption follows a direct local safety path. When an approach sensor detects a train, the owning RL_x begins the warning and gate-protection sequence and sends `CROSSING_STATUS` directly to its two adjacent L_x controllers. Each receiving intersection controller completes all applicable minimum-green, yellow, all-red and pedestrian-clearance requirements before suppressing the movement toward the crossing. After the crossing reopens, the controller applies the bounded queue-drain rule defined in **CC-03**.

The supervisory path is independent of this safety exchange. All nine local controllers send heartbeat, state and fault information to C1, while C1 distributes validated mode, timing-profile and override requests. Each receiving controller retains final local authority and returns `ACK` or `NACK`. If Central becomes unavailable, local controllers enter `DEGRADED_LOCAL`, retain the last validated timing parameters and continue selecting Peak or Off-Peak operation using their local clocks.

Note: To view the diagrams above in greater detail, please access the [A3_report_taskarchitectureddiagram](#).

5.2 Inter-process Communication (IPC) Overview

5.2.1 Internal Process Architecture per Node

Class A: Local Intersection Controller Nodes (L1 to L6 / proc_L1 to proc_L6)

Each intersection controller runs the following dedicated tasks and a separate local watchdog process:

Table 20. Local Intersection Controller Nodes

Sensor_Input_Task	Detects vehicle presence at the stop line, queue warnings from the separate advance detector, and pedestrian push-button events. It filters input noise and forwards validated events to the local traffic controller. Stop-line presence supports <code>OFF_PEAK_SENSOR</code> , while the advance detector is used only for railway-congestion suppression and recovery.
Local_Clock_Timer_Task	Generates local timing events and maintains the time-of-day clock. If Central becomes unavailable, the local clock continues selecting <code>PEAK_FIXED</code> or <code>OFF_PEAK_SENSOR</code> using the last validated timing profile.
Traffic_FSM_Task	Runs the local traffic and pedestrian signal sequences, including the 90 s <code>PEAK_FIXED</code> cycle, bounded <code>OFF_PEAK_SENSOR</code> extensions, required yellow and all-red clearances, pedestrian-request latching, directional railway pre-emption, arterial coordination offsets, and the bounded post-opening queue-drain phase.
Signal_Driver_Task	Drives the vehicle and pedestrian signal heads. It includes a conflict interlock that prevents incompatible movements from receiving green simultaneously and applies to the documented road safe state during a local fault.
Command_Preempt_Comm_Task	Receives <code>CROSSING_STATUS</code> directly from the relevant railway controller. It also receives <code>SET_MODE</code> , <code>SET_TIMING_PROFILE</code> , and <code>REQUEST_OVERRIDE</code> commands from Central, validates each request against local safety conditions, and returns <code>ACK</code> or <code>NACK</code> .
Status_Heartbeat_Comm_Task	Reports the current mode, signal state, vehicle demand, queue warnings, pedestrian requests, configuration version, and local faults to Central. It sends a heartbeat every 1 s and enters <code>DEGRADED_LOCAL</code> after three consecutive missed <code>HEARTBEAT_ACK</code> responses.
Local_Watchdog_Process	Independently supervises the intersection-controller process. If that process stops responding, the watchdog forces the signal outputs to the documented road safe state and reports the fault when communication remains available.

Class B: Railway Level Crossing Safety Controller Nodes (RC1 to RC3 / proc_RC1 to proc_RC3)

Railway-controller nodes govern the safety-critical equipment at physical crossings RC1–RC3:

Table 21. Railway Level-Crossing Controller Nodes

Train_Sensor_Input_Task	Captures and validates train-approach sensor events from both directions and forwards them to the local railway-protection FSM.
Safety_Timer_Task	Generates warning, gate-movement, and train-occupancy timing events. It maintains all active occupancy windows, including overlapping windows when another train is detected before the previous train has cleared.
Railway_Protection_FSM_Task	Manages the crossing sequence activates the warning flashers, lowers the boom gates, verifies both closed-position sensors, permits the train signal to show PROCEED only after closed confirmation, maintains active occupancy windows, and reopens the crossing only after all windows have elapsed. Missing or contradictory gate confirmation causes a local fault-safe response.
Boom_Gate_Driver_Task	Controls the two boom-gate actuators and receives their position-sensor feedback. Elapsed closing time alone is not accepted as confirmation that the gates are closed.
Warning_Train_Signal_Driver_Task	Controls the road-facing flashing lights and train-facing STOP/PROCEED signals. A railway equipment fault causes the train signal to be set to STOP locally without waiting for Central.
Railway_Status_Comm_Task	Send CROSSING_STATUS directly to the two adjacent intersection controllers. It also receives high-level Central requests, such as a railway fault-clear request, validates them locally, and returns ACK or NACK. It does not accept raw gate, flasher, or train-signal commands.
Status_Heartbeat_Comm_Task	Reports crossing state, boom-gate positions, flasher state, train-signal state, active occupancy windows, and equipment faults to Central. It sends a heartbeat every 1 s and enters DEGRADED_LOCAL after three consecutive missed HEARTBEAT_ACK responses.
Local_Watchdog_Process	Independently supervises the railway-controller process. If that process stops responding, the watchdog forces the available railway outputs toward their documented safe state.

Class C: Central Supervisory Controller Node (Central_Controller / proc_central)

The Central Controller coordinates the entire network, aggregates telemetry, and hosts the operator display:

Table 22. Central Controller Nodes

Central_Server_Task	Provides the Central Qnet message endpoint using name_attach(), MsgReceive(), and MsgReply(). It receives heartbeat, status, fault, and reconnection state messages from all nine local controllers and handles command replies.
Supervisor_Mode_Engine_Task	Manages the configurable Peak/Off-Peak schedule, validated timing profiles, and arterial coordination offsets. It sends SET_MODE, SET_TIMING_PROFILE, and bounded REQUEST_OVERRIDE commands to intersection controllers and high-level fault-clear requests to railway controllers.
Watchdog_Monitor_Task	Monitors heartbeat and communication status for all nine local controllers. It raises a communication alarm when a controller becomes unreachable; it does not replace each local node's independent watchdog process.
HMI_Display_Task	Displays current traffic signals, pedestrian requests, vehicle demand, queue warnings, railway-crossing state, equipment faults, controller connectivity, and Central-command outcomes.
Blackbox_Logger_Task	Records timestamped state changes, pre-emption events, mode changes, Central commands, ACK/NACK outcomes, communication failures, and equipment faults in persistent storage.

5.2.2 Inter-Node Message Pathways

IPC-01 - MSG_CROSSING_STATUS_UPDATE

- **Source Task:** `Railway_Status_Comm_Task` (RL1-RL3)
- **Destination Task:** `Command_Preempt_Comm_Task` on the two adjacent Lx controllers
- **Transport Scope:** Direct peer-to-peer RLx → adjacent Lx
- **IPC Mechanism:** `MsgSend()` over Qnet; an urgent approach event may additionally use `MsgSendPulse()`
- **Purpose:** Reports WARNING, CLOSING, CLOSED, TRAIN_PRESENT, OPENING, OPEN, or FAULT.

IPC-02 - MSG_ROAD_STATUS_UPDATE

- **Source Task:** `Status_Heartbeat_Comm_Task` (L1-L6)
- **Destination Task:** `Central_Server_Task`
- **Transport Scope:** L1-L6 → C1
- **IPC Mechanism:** `MsgSend()` / `MsgReply()`
- **Purpose:** Reports mode, traffic phase, signal aspects, demand, queue warnings, pedestrian requests, configuration version, and faults.

IPC-03 - MSG_RAIL_STATUS_UPDATE

- **Source Task:** `Status_Heartbeat_Comm_Task` (RL1-RL3)
- **Destination Task:** `Central_Server_Task`
- **Transport Scope:** RL1-RL3 → C1
- **IPC Mechanism:** `MsgSend()` / `MsgReply()`
- **Purpose:** Reports crossing state, gate positions, flasher state, train-signal state, active occupancy windows, and faults.

IPC-04 - MSG_HEARTBEAT / MSG_HEARTBEAT_ACK

- **Source Task:** `Status_Heartbeat_Comm_Task` on every local controller.
- **Destination Task:** `Central_Server_Task`, with a reply to the sender
- **Transport Scope:** Local controllers ↔ C1
- **IPC Mechanism:** `MsgSend()` / `MsgReply()`
- **Purpose:** Detects Central-link availability at 1 s intervals. Three consecutive missed HEARTBEAT_ACK responses trigger DEGRADED_LOCAL.

IPC-05 - MSG_SET_OPERATION_MODE

- **Source Task:** `Supervisor_Mode_Engine_Task`
- **Destination Task:** `Command_Preempt_Comm_Task` on the target Lx
- **Transport Scope:** C1 → Lx
- **IPC Mechanism:** `MsgSend()` / `MsgReply()`
- **Purpose:** Requests PEAK_FIXED or OFF_PEAK_SENSOR. An accepted change is applied at a safe phase boundary.

IPC-06 - MSG_SET_TIMING_PROFILE

- **Source Task:** `Supervisor_Mode_Engine_Task`
- **Destination Task:** `Command_Preempt_Comm_Task` on the target Lx
- **Transport Scope:** C1 → Lx
- **IPC Mechanism:** `MsgSend()` / `MsgReply()`
- **Purpose:** Supplies validated timing limits, fixed-mode splits, sensor-extension parameters, and the applicable arterial coordination offset.

IPC-07 - MSG_REQUEST_OVERRIDE

- **Source Task:** `Supervisor_Mode_Engine_Task`
- **Destination Task:** `Command_Preempt_Comm_Task` on the target Lx
- **Transport Scope:** C1 → Lx
- **IPC Mechanism:** `MsgSend()` / `MsgReply()`

- **Purpose:** Requests/renews a bounded CLEAR_ROUTE override. The local controller may accept/safely defer/reject it.

IPC-08 - MSG_RAIL_FAULT_CLEAR_REQUEST

- **Source Task:** Supervisor_Mode_Engine_Task
- **Destination Task:** Railway_Status_Comm_Task on the target RLx
- **Transport Scope:** C1 → RLx
- **IPC Mechanism:** MsgSend() / MsgReply()
- **Purpose:** Requests local verification and clearance of a railway-equipment fault after physical repair. It does not directly actuate railway equipment.

IPC-09 - MSG_ACK_REPLY / MSG_NACK_REPLY

- **Source Task:** Receiving command task on Lx or RLx
- **Destination Task:** Central_Server_Task
- **Transport Scope:** Local controller → C1
- **IPC Mechanism:** MsgReply()
- **Purpose:** Reports whether a Central command was accepted, accepted for safe deferral, or rejected, together with the applicable reason code.

IPC-10 - MSG_FULL_STATE_SYNC

- **Source Task:** Reconnected Status_Heartbeat_Comm_Task
- **Destination Task:** Central_Server_Task
- **Transport Scope:** Reconnected local controller → C1
- **IPC Mechanism:** MsgSend() / MsgReply()
- **Purpose:** Sends the controller's current actual state and active configuration before it accepts new Central commands.

5.2.3 Detailed Message Catalog

This catalog provides an itemized breakdown of each inter-node message and its payload structure. Fixed-width integer types are used for portability. Final message sizes and structure alignment must be verified against the implemented C definitions before exact byte totals are stated.

Direct Railway-Crossing Status Message (MSG_CROSSING_STATUS_UPDATE)

- **Message Type:** QNX message sent over a direct Qnet peer channel. An urgent train-approach notification may additionally use a high-priority pulse.
- **Source and Destination:** Sent by Railway_Status_Comm_Task on RL1-RL3 directly to the two adjacent intersection controllers.
- **Operational Purpose:** Reports the complete railway-crossing state so the adjacent controllers can suppress the toward-crossing movement, complete the required safe clearance, and resume traffic after the crossing reports OPEN.
- **Message Content Outline:**

Table 23. Direct Railway-Crossing Status Message Payload (MSG_CROSSING_STATUS_UPDATE)

Field	Data type	Description
header	msg_header_t	Common header containing the message type, source controller, destination controller, sequence number, and timestamp.
crossing_id	uint8_t	Identifies the originating physical crossing: RC1, RC2, or RC3.
crossing_state	uint8_t	Current crossing state: OPEN, WARNING, CLOSING, CLOSED, TRAIN_PRESENT, OPENING, or FAULT.
approach_direction	uint8_t	Identifies the direction from which the train is approaching.
active_occupancy_windows	uint8_t	Number of train-occupancy windows that have not yet expired. This can exceed one when another train is detected before the previous train has cleared.
latest_clear_deadline_ms	uint32_t	Remaining time until the latest active occupancy window expires, in milliseconds.
equipment_fault_flags	uint16_t	Diagnostic bit mask indicating gate-position, actuator, flasher, train-signal, or approach-sensor faults.

This message does not assign a fixed track-drain duration. After receiving OPEN, each adjacent Lx independently applies the CC-03 recovery rule: the connector drain phase extends in 4 s increments while QUEUE_WARNING remains active, up to the 60 s cap.

Road Intersection Status Update (MSG_ROAD_STATUS_UPDATE)

- **Message Type:** QNX synchronous status message over Qnet.
- **Source and Destination:** Sent by Status_Heartbeat_Comm_Task on L1-L6 to Central_Server_Task.
- **Operational Purpose:** Reports the intersection controller's actual traffic, pedestrian, demand, queue, configuration, and fault state to Central.
- **Message Content Outline:**

Table 24. Road Intersection Status Update Payload (MSG_ROAD_STATUS_UPDATE)

Field	Data type	Description
header	msg_header_t	Common header containing the message type, source controller, destination controller, sequence number, and timestamp.
controller_id	uint8_t	Identifies the reporting intersection controller: L1-L6.
normal_mode	uint8_t	Current normal time-of-day mode: PEAK_FIXED or OFF_PEAK_SENSOR.
controller_state	uint8_t	Current controller state or active overlay: NORMAL, RAILWAY_PREEMPTION, CENTRAL_OVERRIDE, DEGRADED_LOCAL, or FAULT_SAFE.
active_phase	uint8_t	Currently active traffic phase: ARTERIAL, CONNECTOR, or an applicable clearance state.
arterial_light	uint8_t	Current arterial signal aspect: RED, YELLOW, or GREEN.
connector_light	uint8_t	Current connector-road signal aspect: RED, YELLOW, or GREEN.
pedestrian_state_mask	uint8_t	Bit mask representing the displayed pedestrian state for the four crossing sides.
demand_present_mask	uint8_t	Bit mask identifying approaches with active stop-line vehicle demand.
queue_warning_mask	uint8_t	Bit mask identifying advance detectors currently reporting QUEUE_WARNING.
pedestrian_request_mask	uint8_t	Bit mask identifying crossing sides with a latched pedestrian request.
phase_remaining_ms	uint32_t	Estimated time remaining in the current timed phase or clearance interval, in milliseconds.
hardware_fault_flags	uint16_t	Diagnostic bit mask for detected input, signal-output, conflict-interlock, or controller faults.
configuration_version	uint32_t	Version number of the timing profile currently validated and active at the controller.

Railway-Crossing Status Update (MSG_RAIL_STATUS_UPDATE)

- **Message Type:** QNX synchronous status message over Qnet.
- **Source & Destination:** Sent by Status_Heartbeat_Comm_Task on RL1-RL3 to Central_Server_Task.
- **Operational Purpose:** Reports on the crossing state and the actual state of locally owned railway equipment.
- **Message Content Outline:**

Table 25. Railway-Crossing Status Update Payload (MSG_RAIL_STATUS_UPDATE)

Field	Data type	Description
header	msg_header_t	Common header containing the message type, source controller, destination controller, sequence number, and timestamp.
crossing_id	uint8_t	Identifies the reporting physical crossing: RC1, RC2, or RC3.
crossing_state	uint8_t	Current crossing state: OPEN, WARNING, CLOSING, CLOSED, TRAIN_PRESENT, OPENING, or FAULT.

gate_1_state	uint8_t	Physical state of the first boom gate: UP, DESCENDING, DOWN, ASCENDING, or FAULT.
gate_2_state	uint8_t	Physical state of the second boom gate using the same state enumeration.
flasher_state_mask	uint8_t	Bit mask indicating which road-facing warning flashers are active or faulted.
train_signal_state_mask	uint8_t	Bit mask representing the STOP, PROCEED, or fault state of the train signals.
active_occupancy_windows	uint8_t	Number of train-occupancy windows that have not yet expired.
latest_clear_deadline_ms	uint32_t	Remaining time until the latest active occupancy window expires, in milliseconds.
hardware_fault_flags	uint16_t	Diagnostic bit mask for gate-position, actuator, flasher, train-signal, or sensor faults.

Heartbeat Message (MSG_HEARTBEAT / MSG_HEARTBEAT_ACK)

- **Message Type:** QNX synchronous message sent every 1 s.
- **Source and Destination:** Sent by each local controller to C1; acknowledged by Central_Server_Task.
- **Operational Purpose:** Detects loss of the Central link.
- **Message Content Outline:**

Table 26. Heartbeat Message Payload (MSG_HEARTBEAT / MSG_HEARTBEAT_ACK)

Field	Data type	Description
header	msg_header_t	Common header containing the message type, source controller, destination controller, sequence number, and timestamp.
controller_state	uint8_t	Summarizes the sender's current state, such as NORMAL, PREEMPTION, OVERRIDE, DEGRADED_LOCAL, or FAULT_SAFE.
configuration_version	uint32_t	Identifies the timing or configuration profile currently active at the sender.
fault_summary	uint16_t	Compact bit mask indicating whether the sender currently has communication, input, output, or controller faults.

Three consecutive missed HEARTBEAT_ACK responses cause the local controller to enter DEGRADED_LOCAL.

Mode Configuration Command (MSG_SET_OPERATION_MODE)

- **Message Type:** QNX synchronous supervisory command.
- **Source and Destination:** Sent by C1 to the target intersection controller.
- **Operational Purpose:** Requests PEAK_FIXED or OFF_PEAK_SENSOR.
- **Message Content Outline:**

Table 27. Mode Configuration Command Payload (MSG_SET_OPERATION_MODE)

Field	Data type	Description
header	msg_header_t	Common header containing the command type, source, destination, sequence number, and timestamp.
target_mode	uint8_t	Requested normal mode: PEAK_FIXED or OFF_PEAK_SENSOR.
apply_at_safe_boundary	uint8_t	Boolean flag requiring the accepted mode change to wait until the current phase and required clearances have completed.
configuration_version	uint32_t	Identifies the configuration version associated with the requested mode.

The receiving controller returns ACK_ACCEPTED_PENDING or NACK. It does not change mode in the middle of a required phase or clearance.

Timing Profile Command (MSG_SET_TIMING_PROFILE)

- **Message Type:** QNX synchronous supervisory command.

- **Source and Destination:** Sent by C1 to the target intersection controller.
- **Operational Purpose:** Supplies validated signal-timing and coordination parameters.
- **Message Content Outline:**

Table 28. Timing Profile Command Payload (MSG_SET_TIMING_PROFILE)

Field	Data type	Description
header	msg_header_t	Common header containing the command type, source, destination, sequence number, and timestamp.
configuration_version	uint32_t	Version assigned to the supplied timing profile.
minimum_green_ms	uint32_t	Minimum permitted green duration; configured as 8,000 ms.
maximum_off_peak_green_ms	uint32_t	Maximum green duration in OFF_PEAK_SENSOR; configured as 40,000 ms.
yellow_ms	uint32_t	Required yellow interval; configured as 4,000 ms.
all_red_ms	uint32_t	Required all-red clearance interval; configured as 2,000 ms.
peak_arterial_green_ms	uint32_t	Arterial green duration in PEAK_FIXED; configured as 48,000 ms.
peak_connector_green_ms	uint32_t	Connector green duration in PEAK_FIXED; configured as 30,000 ms.
sensor_extension_ms	uint32_t	Green-extension increment used in sensor-driven operation; configured as 4,000 ms.
green_wave_offset_ms	uint32_t	Location-specific delay applied to coordinate the intersection with its arterial chain.
apply_at_safe_boundary	uint8_t	Boolean flag requires the validated profile to be applied only at an appropriate safe boundary.

The receiving controller validates every parameter before returning ACK or NACK.

Clear-Route Override Request (MSG_REQUEST_OVERRIDE)

- **Message Type:** QNX synchronous supervisory command.
- **Source and Destination:** Sent by C1 to the target intersection controller.
- **Operational Purpose:** Requests or renews a bounded CLEAR_ROUTE override.
- **Message Content Outline:**

Table 29. Clear-Route Override Request Payload (MSG_REQUEST_OVERRIDE)

Field	Data type	Description
header	msg_header_t	Common header containing the command type, source, destination, sequence number, and timestamp.
override_type	uint8_t	Identifies the requested override operation, currently CLEAR_ROUTE.
target_movement	uint8_t	Identifies the through-movement that Central requests the local controller to prioritize.
requested_duration_ms	uint32_t	Requested override duration in milliseconds, capped at 300,000 ms.
renewal	uint8_t	Boolean flag indicating whether the command is an initial request or a renewal of an active override.

Every request and renewal is validated against pedestrian clearance, railway pre-emption, local faults, and the signal conflict matrix.

Railway Fault-Clear Request (MSG_RAIL_FAULT_CLEAR_REQUEST)

- **Message Type:** QNX synchronous supervisory command.
- **Source and Destination:** Sent by C1 to the target RLX.
- **Operational Purpose:** Requests clearance of a reported railway fault after physical repair.
- **Message Content Outline:**

Table 30. Railway Fault-Clear Request Payload (MSG_RAIL_FAULT_CLEAR_REQUEST)

Field	Data type	Description
header	msg_header_t	Common header containing the command type, source, destination, sequence number, and timestamp.
crossing_id	uint8_t	Identifies the target physical crossing: RC1, RC2, or RC3.
fault_code	uint16_t	Identify the reported railway-equipment fault that the operator requests to clear.
operator_confirmation_id	uint32_t	Correlates the request with the operator's recorded confirmation that physical repair has been completed.

The request does not directly command any railway actuator. RLx independently verifies the physical condition before returning ACK.

Standard Acknowledgement Reply (MSG_ACK_REPLY / MSG_NACK_REPLY)

- **Message Type:** Standard QNX synchronous reply returned using `MsgReply()`.
- **Source and Destination:** Sent by the receiving local controller to C1.
- **Operational Purpose:** Reports the outcome of a Central command.
- **Message Content Outline:**

Table 31. Standard Acknowledgement Reply Payload (MSG_ACK_REPLY / MSG_NACK_REPLY)

Field	Data type	Description
ack_sequence	uint32_t	Matches the reply to the sequence number of the original Central command.
result	uint8_t	Command outcome: ACCEPTED, ACCEPTED_PENDING, REJECTED, or ERROR.
reason_code	uint16_t	Explains a rejection, deferral, validation failure, or processing error. It is zero when no reason code is required.
configuration_version	uint32_t	Reports the configuration version currently active at the local controller after processing the request.

An ACK confirms acceptance or safe deferral; it does not bypass subsequent local safety validation.

Reconnection State Synchronization (MSG_FULL_STATE_SYNC)

- **Message Type:** QNX synchronous state-transfer message.
- **Source and Destination:** Sent by a reconnected local controller to C1.
- **Operational Purpose:** Restores Central's view using the local controller's current actual state before new Central commands are accepted.
- **Message Content Outline:**

Table 32. Reconnection State Synchronisation Payload (MSG_FULL_STATE_SYNC)

Field	Data type	Description
header	msg_header_t	Common header containing the message type, source controller, destination controller, sequence number, and timestamp.
controller_type	uint8_t	Identifies whether the sender is an intersection controller (Lx) or railway controller (RLx).
current_operating_state	uint8_t	Reports the sender's actual current state at the time of reconnection.
configuration_version	uint32_t	Identifies the last validated configuration currently active at the sender.
current_state_payload	Road or railway status structure	Contains the complete applicable MSG_ROAD_STATUS_UPDATE or MSG_RAIL_STATUS_UPDATE state payload.
fault_summary	uint16_t	Summarises faults that remained active or occurred while Central was unreachable.

Central updates its display from this local state before sending new commands. It does not overwrite the controller with a stale pre-disconnection state.

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7. Appendices

Appendix A: System Component Nomenclature and Mapping

This appendix establishes the standardized nomenclature, operational identifiers, and functional mapping for all physical infrastructure, logical controllers, and hardware actuators within the distributed traffic-control network. These definitions ensure exact consistency between the physical intersection layouts and the software implementation architecture.

A1. Physical Locations and Road Network

Table 33. Physical Locations and Road Network Identifiers

<u>Identifier</u>	<u>Category</u>	<u>System Description</u>
I1–I6	Physical road intersections	The six signalized intersections are shown in the supplied network topology. I _x identifies a physical location, not a controller or QNX node.
RC1–RC3	Physical railway crossings	The three locations where the double-track railway crosses R3, R4, and R5, respectively.
R1–R2	Arterial roads	The two continuous arterial routes: R1 links I1–I3–I5, while R2 links I2–I4–I6. These roads receive higher base priority and arterial coordination.
R3–R5	Connector roads	The three connector roads linking I1–I2, I3–I4, and I5–I6, respectively. Each connector crosses the railway once, at RC1, RC2, or RC3.

A2. Logical Controllers

The system contains ten logical controllers: one Central Controller, six intersection local controllers, and three railway-crossing local controllers. A logical controller is not necessarily equivalent to one QNX node or one physical computer. The deployment of controller processes across multiple QNX nodes is defined separately in **Section 5.1**, Task Architecture Diagram.

Table 34. Logical Controller Deployment and Mapping

<u>Controller ID</u>	<u>Controller Type</u>	<u>Physical Location</u>
C1	Central Controller	Central control room
L1–L6	Intersection local controllers	L _x controls physical intersection I _x
RL1	Railway-crossing local controller	Controls RC1 on R3
RL2	Railway-crossing local controller	Controls RC2 on R4
RL3	Railway-crossing local controller	Controls RC3 on R5

All three railway-crossing controllers use the same control logic and operate independently. No railway controller commands an intersection controller or another railway controller. Each RL_x only supplies crossing-status information to the two adjacent L_x controllers, which then determine their own safe traffic response.

A3. Intersection Sensors and Actuators

Table 35. Intersection Sensor and Actuator Nomenclature

<u>Identifier</u>	<u>Component</u>	<u>Ownership and Function</u>
VD_ _x	Vehicle-presence detector	One approach-level binary presence sensor for each of the four approaches at an intersection. It reports DEMAND_PRESENT or DEMAND_ABSENT to the owning L _x and supports OFF_PEAK_SENSOR operation. It does not count vehicles or distinguish their intended turning movement.
QD_ _x	Advance/queue detector	A binary detector positioned approximately 60 - 80 m behind the stop line on an approach that feeds directly toward a railway crossing. It indicates that a queue has reached the detection point and supports congestion-suppression logic. It is included only in the relevant crossing-facing approaches.
PB_ _x	Pedestrian push-button	One latched request input for each of the four pedestrian crossing sides at an intersection. A press records a pending request until the compatible pedestrian phase is served.
VS_ _x	Vehicle signal head	One circular vehicle signal head for each approach, controlled exclusively by the owning L _x . Its normal states are RED, YELLOW, and GREEN; its defined safe fallback state is FLASHING_RED.

PS_x	Pedestrian signal head	One pedestrian signal for each crossing side, controlled exclusively by the owning L _x . Its states are WALK, FLASHING DONT WALK, and DONT WALK.
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A4. Railway-Crossing Sensors and Actuators

Table 36. Railway-Crossing Sensors and Actuator Nomenclature

Identifier	Component	Ownership and Function
TS_x	Train-approach sensor	One upstream sensor for each rail direction at a crossing. It reports TRAIN_APPROACHING to the owner RL _x and initiates the local railway safety sequence. Its placement is based on the assumed train speed and the approximately 45 s warning-to-arrival requirement derived in Appendix B; it does not inherently guarantee an exact detection time.
GS_x	Boom-gate position sensor	One position-sensing mechanism for each boom gate. It reports whether the gate is confirmed open, confirmed closed, moving, or faulty. Elapsed closing time alone is never treated as proof that a gate is safely closed.
BG_x	Boom gate	A physical road barrier controlled exclusively by the owning RL _x . Each railway crossing has two boom gates. Its states are UP, MOVING, DOWN, and FAULT.
FL_x	Road-facing flashing light	A red warning beacon controlled exclusively by the owning RL _x . Each crossing has two road-facing flasher sets, which activate before the boom gates begin closing and remain active throughout the closure.
RS_x	Railway signal	A signal facing approaching trains, controlled exclusively by the owning RL _x . Each crossing has one signal for each rail direction. PROCEED is permitted only after the relevant boom gates are confirmed to be closed; any gate or crossing fault forces the railway signal to STOP.

A5. Identifier Qualification

The suffix *x* represents the owning location, approach, crossing side, gate, or rail direction as appropriate. Full implementation identifiers must include enough information to uniquely identify both the owning controller and the physical component. Examples include:

- L1.VD_N : north-approach vehicle detector owned by L1
- L1.PB_E : east-side pedestrian button owned by L1
- RL1.TS_EB : eastbound train-approach sensor owned by RL1
- RL1.BG_A : boom gate A at RC1
- RL1.RS_EB : eastbound railway signal at RC1

The final source-code constants and IPC message fields must use this nomenclature consistently.

A6. Mapping Summary

The principal controller-to-location mapping is:

- C1 monitors and coordinates the distributed system but owns no physical traffic or railway actuator.
- L1–L6 exclusively control the road and pedestrian signals at I1–I6.
- RL1–RL3 exclusively control the railway equipment at RC1–RC3.
- RL1 shares RC1 status with L1 and L2.
- RL2 shares RC2 status with L3 and L4.
- RL3 shares RC3 status with L5 and L6.

This mapping separates physical locations from logical controllers and does not prescribe a one-controller-to-one-QNX-node or one-controller-to-one-computer deployment.

Appendix B: Supporting Assumption Derivations

This appendix provides mathematical derivations and numeric justifications underpinning the core traffic and safety assumptions. These detailed calculations validate the operational safety margins, signal timings, and coordination offsets applied throughout the network architecture.

B1. Peak Fixed-Cycle Timing

Table 37. Peak Fixed-Cycle Phase Allocation and Clearance Timing

<u>Phase</u>	<u>Green / s</u>	<u>Yellow / s</u>	<u>Red clearance / s</u>	<u>Phase allocation / s</u>
Arterial	48	4	2	54
Connector	30	4	2	36
Total cycle				90

The arterial receives 18 seconds more green than the connector, implementing the qualitative priority in **DP-04**. These are tunable configuration values rather than measured site timings.

B2. Arterial Green-Wave Offsets

The calculation uses $\text{offset} = \text{distance} \div 16.67 \text{ m/s}$, based on **TC-01**.

Table 38. Arterial Green-Wave Coordination Offsets

<u>Segment</u>	<u>Distance / m</u>	<u>Segment offset / s</u>	<u>Cumulative offset from chain start / s</u>
L1 → L3 on R1	350	21	21
L3 → L5 on R1	400	24	45
L2 → L4 on R2	320	19	19
L4 → L6 on R2	380	23	42

B3. Congestion Drain Rule

Table 39. Congestion Drain Phase Parameters

<u>Input or limit</u>	<u>Value</u>
Queue input	Binary QUEUE_WARNING from CC-01
Recheck/extension increment	4 s
Maximum drain green	60 s
Termination	QUEUE_WARNING clears or the 60 seconds cap is reached

The binary detector does not support a queue-length formula. Drain time is therefore selected by repeated state checks within a bounded interval.

B4. Railway Warning Lead Time and Sensor Placement

Table 40. Railway Warning Lead Time Budget Calculation

<u>Component</u>	<u>Duration / s</u>	<u>Running total / s</u>
Flashing-only warning	5	5
Gate-closing allowance	10	15
Closed-confirmation margin	5	20
Adjacent-intersection clearance allowance	25	45

Using the **RC-08** maximum train speed:

$$\text{Approach distance} = 45 \text{ s} \times 22.2 \text{ m/s} \approx 999 \text{ m} \approx 1,000 \text{ m}$$

The timing and placement values are tunable PoC assumptions. Actual deployment would require a site-specific railway and traffic engineering study.